PART X

Automorphisms and Galois Theory

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SECTION 48 AUTOMORPHISMS OF FIELDS

The Conjugation Isomorphisms of Algebraic Field Theory

Let F be a field, and let $\bar{F}$ be an algebraic closure of F, that is, an algebraic extension of F that is algebraically closed. Such a field $\bar{F}$ exists, by Theorem 31.17. Our selection of a particular $\bar{F}$ is not critical, since, as we shall show in Section 49, any two algebraic closures of F are isomorphic under a map leaving F fixed. *From now on in our work, we shall assume that all algebraic extensions and all elements algebraic over a field F under consideration are contained in one fixed algebraic closure $\bar{F}$ of F.*

Remember that we are engaged in the study of zeros of polynomials. In the terminology of Section 31, studying zeros of polynomials in $F[x]$ amounts to studying the structure of algebraic extensions of F and of elements algebraic over F. We shall show that if E is an algebraic extension of F with $\alpha, \beta \in E$, then α and β have the same algebraic properties if and only if $\text{irr}(\alpha, F) = \text{irr}(\beta, F)$. We shall phrase this fact in terms of mappings, as we have been doing all along in field theory. We achieve this by showing that if $\text{irr}(\alpha, F) = \text{irr}(\beta, F)$, then there exists an isomorphism $\psi_{\alpha,\beta}$ of $F(\alpha)$ onto $F(\beta)$ that maps each element of F onto itself and maps α onto β. The next theorem exhibits this isomorphism $\psi_{\alpha,\beta}$. These isomorphisms will become our fundamental tools for the study of algebraic extensions; they supplant the *evaluation homomorphisms* ϕ_α of Theorem 22.4, which make their last contribution in defining these isomorphisms. Before stating and proving this theorem, let us introduce some more terminology.

† Section 52 is not required for the remainder of the text.

48.1 Definition Let E be an algebraic extension of a field F. Two elements $\alpha, \beta \in E$ are **conjugate over** F if $\text{irr}(\alpha, F) = \text{irr}(\beta, F)$, that is, if α and β are zeros of the same irreducible polynomial over F. ■

48.2 Example The concept of conjugate elements just defined conforms with the classic idea of *conjugate complex numbers* if we understand that by conjugate complex numbers we mean numbers that are *conjugate over* $\mathbb{R}$. If $a, b \in \mathbb{R}$ and $b \neq 0$, the conjugate complex numbers $a + bi$ and $a - bi$ are both zeros of $x^2 - 2ax + a^2 + b^2$, which is irreducible in $\mathbb{R}[x]$. ▲

48.3 Theorem **(The Conjugation Isomorphisms)** Let F be a field, and let α and β be algebraic over F with $\deg(\alpha, F) = n$. The map $\psi_{\alpha,\beta} : F(\alpha) \to F(\beta)$ defined by

$$\psi_{\alpha,\beta}(c_0 + c_i\alpha + \cdots + c_{n-1}\alpha^{n-1}) = c_0 + c_1\beta + \cdots + c_{n-1}\beta^{n-1}$$

for $c_i \in F$ is an isomorphism of $F(\alpha)$ onto $F(\beta)$ if and only if α and β are conjugate over F.

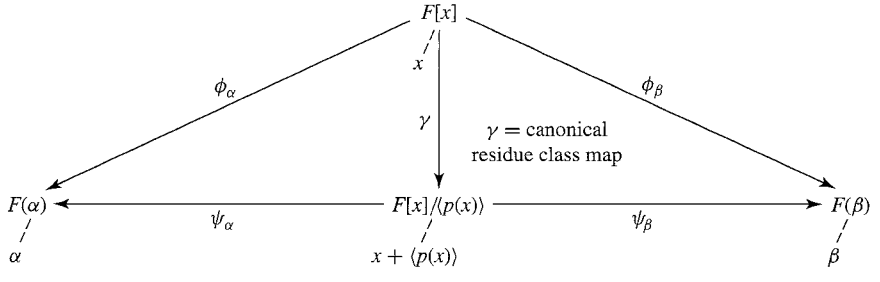
Proof Suppose that $\psi_{\alpha,\beta} : F(\alpha) \to F(\beta)$ as defined in the statement of the theorem is an isomorphism. Let $\text{irr}(\alpha, F) = a_0 + a_1x + \cdots + a_nx^n$. Then $a_0 + a_1\alpha + \cdots + a_n\alpha^n = 0$, so

$$\psi_{\alpha,\beta}(a_0 + a_1\alpha + \cdots + a_n\alpha^n) = a_0 + a_1\beta + \cdots + a_n\beta^n = 0.$$

By the last assertion in the statement of Theorem 29.13 this implies that $\text{irr}(\beta, F)$ divides $\text{irr}(\alpha, F)$. A similar argument using the isomorphism $(\psi_{\alpha,\beta})^{-1} = \psi_{\beta,\alpha}$ shows that $\text{irr}(\alpha, F)$ divides $\text{irr}(\beta, F)$. Therefore, since both polynomials are monic, $\text{irr}(\alpha, F) = \text{irr}(\beta, F)$, so α and β are conjugate over F.

Conversely, suppose $\text{irr}(\alpha, F) = \text{irr}(\beta, F) = p(x)$. Then the evaluation homomorphisms $\phi_\alpha : F[x] \to F(\alpha)$ and $\phi_\beta : F[x] \to F(\beta)$ both have the same kernel $\langle p(x)\rangle$. By Theorem 26.17, corresponding to $\phi_\alpha : F[x] \to F(\alpha)$, there is a natural isomorphism ψ_α mapping $F[x]/\langle p(x)\rangle$ onto $\phi_\alpha[F[x]] = F(\alpha)$. Similarly, ϕ_β gives rise to an isomorphism ψ_β mapping $F[x]/\langle p(x)\rangle$ onto $F(\beta)$. Let $\psi_{\alpha,\beta} = \psi_\beta(\psi_\alpha)^{-1}$. These mappings are diagrammed in Fig. 48.4 where the dashed lines indicate corresponding elements under the mappings. As the composition of two isomorphisms, $\psi_{\alpha,\beta}$ is again an isomorphism and maps $F(\alpha)$ onto $F(\beta)$. For $(c_0 + c_1\alpha + \cdots + c_{n-1}\alpha^{n-1}) \in F(\alpha)$, we have

$$\begin{aligned}&\psi_{\alpha,\beta}(c_0 + c_1\alpha + \cdots + c_{n-1}\alpha^{n-1})\\ &\quad = (\psi_\beta\psi_\alpha{}^{-1})(c_0 + c_1\alpha + \cdots + c_{n-1}\alpha^{n-1})\end{aligned}$$



48.4 Figure

$$= \psi_\beta((c_0 + c_1 x + \cdots + c_{n-1}x^{n-1}) + \langle p(x)\rangle)$$
$$= c_0 + c_1\beta + \cdots + c_{n-1}\beta^{n-1}.$$

Thus $\psi_{\alpha,\beta}$ is the map defined in the statement of the theorem. ◆

The following corollary of Theorem 48.3 is the cornerstone of our proof of the important Isomorphism Extension Theorem of Section 49 and of most of the rest of our work.

48.5 Corollary Let α be algebraic over a field F. Every isomorphism ψ mapping $F(\alpha)$ onto a subfield of $\bar{F}$ such that $\psi(a) = a$ for $a \in F$ maps α onto a conjugate β of α over F. Conversely, for each conjugate β of α over F, there exists exactly one isomorphism $\psi_{\alpha,\beta}$ of $F(\alpha)$ onto a subfield of $\bar{F}$ mapping α onto β and mapping each $a \in F$ onto itself.

Proof Let ψ be an isomorphism of $F(\alpha)$ onto a subfield of $\bar{F}$ such that $\psi(a) = a$ for $a \in F$. Let $\text{irr}(\alpha, F) = a_0 + a_1 x + \cdots + a_n x^n$. Then

$$a_0 + a_1\alpha + \cdots + a_n\alpha^n = 0,$$

so

$$0 = \psi(a_0 + a_1\alpha + \cdots + a_n\alpha^n) = a_0 + a_1\psi(\alpha) + \cdots + a_n\psi(\alpha)^n,$$

and $\beta = \psi(\alpha)$ is a conjugate of α.

Conversely, for each conjugate β of α over F, the conjugation isomorphism $\psi_{\alpha,\beta}$ of Theorem 48.3 is an isomorphism with the desired properties. That $\psi_{\alpha,\beta}$ is the only such isomorphism follows from the fact that an isomorphism of $F(\alpha)$ is completely determined by its values on elements of F and its value on α. ◆

As a second corollary of Theorem 48.3, we can prove a familiar result.

48.6 Corollary Let $f(x) \in \mathbb{R}[x]$. If $f(a + bi) = 0$ for $(a + bi) \in \mathbb{C}$, where $a, b \in \mathbb{R}$, then $f(a - bi) = 0$ also. Loosely, complex zeros of polynomials with real coefficients occur in conjugate pairs.

Proof We have seen that $\mathbb{C} = \mathbb{R}(i)$. Now

$$\text{irr}(i, \mathbb{R}) = \text{irr}(-i, \mathbb{R}) = x^2 + 1,$$

so i and $-i$ are conjugate over $\mathbb{R}$. By Theorem 48.3, the conjugation map $\psi_{i,-i} : \mathbb{C} \to \mathbb{C}$ where $\psi_{i,-i}(a + bi) = a - bi$ is an isomorphism. Thus, if for $a_i \in \mathbb{R}$,

$$f(a + bi) = a_0 + a_1(a + bi) + \cdots + a_n(a + bi)^n = 0,$$

then

$$0 = \psi_{i,-i}(f(a + bi)) = a_0 + a_1(a - bi) + \cdots + a_n(a - bi)^n$$
$$= f(a - bi),$$

that is, $f(a - bi) = 0$ also. ◆

48.7 Example Consider $\mathbb{Q}(\sqrt{2})$ over $\mathbb{Q}$. The zeros of $\text{irr}(\sqrt{2}, \mathbb{Q}) = x^2 - 2$ are $\sqrt{2}$, and $-\sqrt{2}$, so $\sqrt{2}$ and $-\sqrt{2}$ are conjugate over $\mathbb{Q}$. According to Theorem 48.3 the map $\psi_{\sqrt{2},-\sqrt{2}} : \mathbb{Q}(\sqrt{2}) \to \mathbb{Q}(\sqrt{2})$ defined by

$$\psi_{\sqrt{2},-\sqrt{2}}(a + b\sqrt{2}) = a - b\sqrt{2}$$

is an isomorphism of $\mathbb{Q}(\sqrt{2})$ onto itself. ▲

Automorphisms and Fixed Fields

As illustrated in the preceding corollary and example, a field may have a nontrivial isomorphism onto itself. *Such maps will be of utmost importance in the work that follows.*

48.8 Definition An isomorphism of a field onto itself is an **automorphism of the field.** ■

48.9 Definition If σ is an isomorphism of a field E onto some field, then an element a of E is **left fixed by** σ if $\sigma(a) = a$. A collection S of isomorphisms of E **leaves a subfield** F **of** E **fixed** if each $a \in F$ is left fixed by every $\sigma \in S$. If $\{\sigma\}$ leaves F fixed, then σ **leaves** F **fixed.** ■

48.10 Example Let $E = \mathbb{Q}(\sqrt{2}, \sqrt{3})$. The map $\sigma : E \to E$ defined by

$$\sigma(a + b\sqrt{2} + c\sqrt{3} + d\sqrt{6}) = a + b\sqrt{2} - c\sqrt{3} - d\sqrt{6}$$

for $a, b, c, d \in \mathbb{Q}$ is an automorphism of E; it is the conjugation isomorphism $\psi_{\sqrt{3},-\sqrt{3}}$ of E onto itself if we view E as $(\mathbb{Q}(\sqrt{2}))(\sqrt{3})$. We see that σ leaves $\mathbb{Q}(\sqrt{2})$ fixed. ▲

It is our purpose to study the structure of an algebraic extension E of a field F by studying the automorphisms of E that leave fixed each element of F. We shall presently show that these automorphisms form a group in a natural way. We can then apply the results concerning group structure to get information about the structure of our field extension. Thus much of our preceding work is now being brought together. The next three theorems are readily proved, but the ideas contained in them form the foundation for everything that follows. These theorems are therefore of great importance to us. They really amount to observations, rather than theorems; it is the *ideas* contained in them that are important. A big step in mathematics does not always consist of proving a *hard* theorem, but may consist of noticing how certain known mathematics may relate to new situations. Here we are bringing group theory into our study of zeros of polynomials. Be sure to understand the concepts involved. Unlikely as it may seem, they are the key to the solution of our *final goal* in this text.

Final Goal (to be more precisely stated later): To show that not all zeros of every quintic (degree 5) polynomial $f(x)$ can be expressed in terms of radicals starting with elements in the field containing the coefficients of $f(x)$.

Historical Note

It was Richard Dedekind who first developed the idea of an automorphism of a field, what he called a "permutation of the field," in 1894. The earlier application of group theory to the theory of equations had been through groups of permutations of the roots of certain polynomials. Dedekind extended this idea to mappings of the entire field and proved several of the theorems of this section.

Though Heinrich Weber continued Dedekind's approach to groups acting on fields in his algebra text of 1895, this method was not pursued in other texts near the turn of the century. It was not until the 1920s, after Emmy Noether's abstract approach to algebra became influential at Gottingen, that Emil Artin (1898–1962) developed this relationship of groups and fields in great detail. Artin emphasized that the goal of what is now called Galois theory should not be to determine solvability conditions for algebraic equations, but to explore the relationship between field extensions and groups of automorphisms. Artin detailed his approach in a lecture given in 1926; his method was first published in B. L. Van der Waerden's *Modern Algebra* text of 1930 and later by Artin himself in lecture notes in 1938 and 1942. In fact, the remainder of this text is based on Artin's development of Galois theory.

If $\{\sigma_i \mid i \in I\}$ is a collection of automorphisms of a field E, the elements of E about which $\{\sigma_i \mid i \in I\}$ gives the least information are those $a \in E$ left fixed by every σ_i for $i \in I$. This first of our three theorems contains almost all that can be said about these fixed elements of E.

48.11 Theorem Let $\{\sigma_i \mid i \in I\}$ be a collection of automorphisms of a field E. Then the set $E_{\{\sigma_i\}}$ of all $a \in E$ left fixed by every σ_i for $i \in I$ forms a subfield of E.

Proof If $\sigma_i(a) = a$ and $\sigma_i(b) = b$ for all $i \in I$, then

$$\sigma_i(a \pm b) = \sigma_i(a) \pm \sigma_i(b) = a \pm b$$

and

$$\sigma_i(ab) = \sigma_i(a)\sigma_i(b) = ab$$

for all $i \in I$. Also, if $b \neq 0$, then

$$\sigma_i(a/b) = \sigma_i(a)/\sigma_i(b) = a/b$$

for all $i \in I$. Since the σ_i are automorphisms, we have

$$\sigma_i(0) = 0 \qquad \text{and} \qquad \sigma_i(1) = 1$$

for all $i \in I$. Hence $0, 1 \in E_{\{\sigma_i\}}$ Thus $E_{\{\sigma_i\}}$ is a subfield of E. ◆

48.12 Definition The field $E_{\{\sigma_i\}}$ of Theorem 48.11 is the **fixed field of** $\{\sigma_i \mid i \in I\}$. For a single automorphism σ, we shall refer to $E_{\{\sigma\}}$ as the **fixed field of** σ. ■

48.13 Example Consider the automorphism $\psi_{\sqrt{2},-\sqrt{2}}$ of $\mathbb{Q}(\sqrt{2})$ given in Example 48.7. For $a, b \in \mathbb{Q}$, we have

$$\psi_{\sqrt{2},-\sqrt{2}}(a + b\sqrt{2}) = a - b\sqrt{2},$$

and $a - b\sqrt{2} = a + b\sqrt{2}$ if and only if $b = 0$. Thus the fixed field of $\psi_{\sqrt{2},-\sqrt{2}}$ is $\mathbb{Q}$. ▲

Note that an automorphism of a field E is in particular a one-to-one mapping of E onto E, that is, a *permutation of* E. If σ and τ are automorphisms of E, then the permutation $\sigma\tau$ is again an automorphism of E, since, in general, composition of homomorphisms again yields a homomorphism. This is how group theory makes its entrance.

48.14 Theorem The set of all automorphisms of a field E is a group under function composition.

Proof Multiplication of automorphisms of E is defined by function composition, and is thus associative (it is *permutation multiplication*). The identity permutation $\iota : E \to E$ given by $\iota(\alpha) = \alpha$ for $\alpha \in E$ is an automorphism of E. If σ is an automorphism, then the permutation σ^{-1} is also an automorphism. Thus all automorphisms of E form a subgroup of S_E, the group of all permutations of E given by Theorem 8.5. ◆

48.15 Theorem Let E be a field, and let F be a subfield of E. Then the set $G(E/F)$ of all automorphisms of E leaving F fixed forms a subgroup of the group of all automorphisms of E. Furthermore, $F \leq E_{G(E/F)}$.

Proof For $\sigma, \tau \in G(E/F)$ and $a \in F$, we have

$$(\sigma\tau)(a) = \sigma(\tau(a)) = \sigma(a) = a,$$

so $\sigma\tau \in G(E/F)$. Of course, the identity automorphism ι is in $G(E/F)$. Also, if $\sigma(a) = a$ for $a \in F$, then $a = \sigma^{-1}(a)$ so $\sigma \in G(E/F)$ implies that $\sigma^{-1} \in G(E/F)$. Thus $G(E/F)$ is a subgroup of the group of all automorphisms of E.

Since every element of F is left fixed by every element of $G(E/F)$, it follows immediately that the field $E_{G(E/F)}$ of *all* elements of E left fixed by $G(E/F)$ contains F. ◆

48.16 Definition The group $G(E/F)$ of the preceding theorem is the **group of automorphisms of E leaving F fixed**, or, more briefly, the **group of E over F**.

Do not think of E/F in the notation $G(E/F)$ as denoting a quotient space of some sort, but rather as meaning that E is an extension field of the field F.

The ideas contained in the preceding three theorems are illustrated in the following example. We urge you to study this example carefully. ■

48.17 Example Consider the field $\mathbb{Q}(\sqrt{2}, \sqrt{3})$. Example 31.9 shows that $[\mathbb{Q}(\sqrt{2}, \sqrt{3}) : \mathbb{Q}] = 4$. If we view $\mathbb{Q}(\sqrt{2}, \sqrt{3})$ as $(\mathbb{Q}(\sqrt{3}))(\sqrt{2})$, the conjugation isomorphism $\psi_{\sqrt{2},-\sqrt{2}}$ of Theorem 48.3 defined by

$$\psi_{\sqrt{2},-\sqrt{2}}(a + b\sqrt{2}) = a - b\sqrt{2}$$

for $a, b \in \mathbb{Q}(\sqrt{3})$ is an automorphism of $\mathbb{Q}(\sqrt{2}, \sqrt{3})$ having $\mathbb{Q}(\sqrt{3})$ as fixed field. Similarly, we have the automorphism $\psi_{\sqrt{3},-\sqrt{3}}$ of $\mathbb{Q}(\sqrt{2}, \sqrt{3})$ having $\mathbb{Q}(\sqrt{2})$ as fixed field. Since the product of two automorphisms is an automorphism, we can consider $\psi_{\sqrt{2},-\sqrt{2}}\psi_{\sqrt{3},-\sqrt{3}}$ which **moves** both $\sqrt{2}$ and $\sqrt{3}$, that is, leaves neither number fixed. Let

$$\begin{aligned} \iota &= \text{the identity automorphism,} \\ \sigma_1 &= \psi_{\sqrt{2},-\sqrt{2}}, \\ \sigma_2 &= \psi_{\sqrt{3},-\sqrt{3}}, \text{ and} \\ \sigma_3 &= \psi_{\sqrt{2},-\sqrt{2}}\psi_{\sqrt{3},-\sqrt{3}}. \end{aligned}$$

48.18 Table

	ι	σ_1	σ_2	σ_3
ι	ι	σ_1	σ_2	σ_3
σ_1	σ_1	ι	σ_3	σ_2
σ_2	σ_2	σ_3	ι	σ_1
σ_3	σ_3	σ_2	σ_1	ι

The group of all automorphisms of $\mathbb{Q}(\sqrt{2}, \sqrt{3})$ has a fixed field, by Theorem 48.11. This fixed field must contain $\mathbb{Q}$, since every automorphism of a field leaves 1 and hence the prime subfield fixed. A basis for $\mathbb{Q}(\sqrt{2}, \sqrt{3})$ over $\mathbb{Q}$ is $\{1, \sqrt{2}, \sqrt{3}, \sqrt{6}\}$. Since $\sigma_1(\sqrt{2}) = -\sqrt{2}$, $\sigma_1(\sqrt{6}) = -\sqrt{6}$ and $\sigma_2(\sqrt{3}) = -\sqrt{3}$, we see that $\mathbb{Q}$ is exactly the fixed field of $\{\iota, \sigma_1, \sigma_2, \sigma_3\}$. It is readily checked that $G = \{\iota, \sigma_1, \sigma_2, \sigma_3\}$ is a group under automorphism multiplication (function composition). The group table for G is given in Table 48.18. For example,

$$\sigma_1\sigma_3 = \psi_{\sqrt{2},-\sqrt{2}}(\psi_{\sqrt{2},-\sqrt{2}}\psi_{\sqrt{3},-\sqrt{3}}) = \psi_{\sqrt{3},-\sqrt{3}} = \sigma_2.$$

The group G is isomorphic to the Klein 4-group. We can show that G is the full group $\mathbb{G}(\mathbb{Q}(\sqrt{2}, \sqrt{3})/\mathbb{Q})$, because every automorphism τ of $\mathbb{Q}(\sqrt{2}, \sqrt{3})$ maps $\sqrt{2}$ onto either $\pm\sqrt{2}$, by Corollary 48.5. Similarly, τ maps $\sqrt{3}$ onto either $\pm\sqrt{3}$. But since $\{1, \sqrt{2}, \sqrt{3}, \sqrt{2}\sqrt{3}\}$ is a basis for $\mathbb{Q}(\sqrt{2}, \sqrt{3})$ over $\mathbb{Q}$, an automorphism of $\mathbb{Q}(\sqrt{2}, \sqrt{3})$ leaving $\mathbb{Q}$ fixed is determined by its values on $\sqrt{2}$ and $\sqrt{3}$. Now, ι, σ_1, σ_2, and σ_3 give all possible combinations of values on $\sqrt{2}$ and $\sqrt{3}$, and hence are all possible automorphisms of $\mathbb{Q}(\sqrt{2}, \sqrt{3})$.

Note that $G(\mathbb{Q}(\sqrt{2}, \sqrt{3})/\mathbb{Q})$ has order 4, and $[\mathbb{Q}(\sqrt{2}, \sqrt{3}) : \mathbb{Q}] = 4$. *This is no accident*, but rather an instance of a general situation, as we shall see later. ▲

The Frobenius Automorphism

Let F be a finite field. We shall show later that the group of all automorphisms of F is cyclic. Now a cyclic group has by definition a generating element, and it may have several generating elements. For an abstract cyclic group there is no way of distinguishing any one generator as being more important than any other. However, for the cyclic group of all automorphisms of a finite field there is a canonical (natural) generator, the *Frobenius automorphism* (classically, the *Frobenius substitution*). This fact is of considerable importance in some advanced work in algebra. The next theorem exhibits this Frobenius automorphism.

48.19 Theorem Let F be a finite field of characteristic p. Then the map $\sigma_p : F \to F$ defined by $\sigma_p(a) = a^p$ for $a \in F$ is an automorphism, the **Frobenius automorphism,** of F. Also, $F_{\{\sigma_p\}} \simeq \mathbb{Z}_p$.

Proof Let $a, b \in F$. Taking $n = 1$ in Lemma 33.9, we see that $(a + b)^p = a^p + b^p$. Thus we have

$$\sigma_p(a + b) = (a + b)^p = a^p + b^p = \sigma_p(a) + \sigma_p(b).$$

Of course,

$$\sigma_p(ab) = (ab)^p = a^p b^p = \sigma_p(a)\sigma_p(b),$$

so σ_p is at least a homomorphism. If $\sigma_p(a) = 0$, then $a^p = 0$, and $a = 0$, so the kernel of σ_p is $\{0\}$, and σ_p is a one-to-one map. Finally, since F is finite, σ_p is onto, by counting. Thus σ_p is an automorphism of F.

The prime field $\mathbb{Z}_p$ must be contained (up to isomorphism) in F, since F is of characteristic p. For $c \in \mathbb{Z}_p$, we have $\sigma_p(c) = c^p = c$, by Fermat's theorem (see Corollary 20.2). Thus the polynomial $x^p - x$ has p zeros in F, namely the elements of $\mathbb{Z}_p$. By Corollary 23.5, a polynomial of degree n over a field can have at most n zeros in the field. Since the elements fixed under σ_p are precisely the zeros in F of $x^p - x$, we see that

$$\mathbb{Z}_p = F_{\{\sigma_p\}}. \quad \blacklozenge$$

Freshmen in college still sometimes make the error of saying that $(a + b)^n = a^n + b^n$. Here we see that this *freshman exponentiation*, $(a + b)^p = a^p + b^p$ with exponent p, is actually valid in a field F of characteristic p.

EXERCISES 48

Computations

In Exercises 1 through 8, find all conjugates in $\mathbb{C}$ of the given number over the given field.

1. $\sqrt{2}$ over $\mathbb{Q}$

2. $\sqrt{2}$ over $\mathbb{R}$

3. $3 + \sqrt{2}$ over $\mathbb{Q}$

4. $\sqrt{2} - \sqrt{3}$ over $\mathbb{Q}$

5. $\sqrt{2} + i$ over $\mathbb{Q}$

6. $\sqrt{2} + i$ over $\mathbb{R}$

7. $\sqrt{1 + \sqrt{2}}$ over $\mathbb{Q}$

8. $\sqrt{1 + \sqrt{2}}$ over $\mathbb{Q}(\sqrt{2})$

In Exercises 9 through 14, we consider the field $E = \mathbb{Q}(\sqrt{2}, \sqrt{3}, \sqrt{5})$. It can be shown that $[E : \mathbb{Q}] = 8$. In the notation of Theorem 48.3, we have the following conjugation isomorphisms (which are here automorphisms of E):

$$\psi_{\sqrt{2},-\sqrt{2}} : (\mathbb{Q}(\sqrt{3}, \sqrt{5}))(\sqrt{2}) \to (\mathbb{Q}(\sqrt{3}, \sqrt{5}))(-\sqrt{2}),$$
$$\psi_{\sqrt{3},-\sqrt{3}} : (\mathbb{Q}(\sqrt{2}, \sqrt{5}))(\sqrt{3}) \to (\mathbb{Q}(\sqrt{2}, \sqrt{5}))(-\sqrt{3}),$$
$$\psi_{\sqrt{5},-\sqrt{5}} : (\mathbb{Q}(\sqrt{2}, \sqrt{3}))(\sqrt{5}) \to (\mathbb{Q}(\sqrt{2}, \sqrt{3}))(-\sqrt{5}).$$

For shorter notation, let $\tau_2 = \psi_{\sqrt{2},-\sqrt{2}}$, $\tau_3 = \psi_{\sqrt{3},-\sqrt{3}}$, and $\tau_5 = \psi_{\sqrt{5},-\sqrt{5}}$. Compute the indicated element of E.

9. $\tau_2(\sqrt{3})$

10. $\tau_2(\sqrt{2} + \sqrt{5})$

11. $(\tau_3\tau_2)(\sqrt{2} + 3\sqrt{5})$

12. $(\tau_5\tau_3)\left(\dfrac{\sqrt{2} - 3\sqrt{5}}{2\sqrt{3} - \sqrt{2}}\right)$

13. $(\tau_5{}^2\tau_3\tau_2)(\sqrt{2} + \sqrt{45})$

14. $\tau_3[\tau_5(\sqrt{2} - \sqrt{3} + (\tau_2\tau_5)(\sqrt{30}))]$

15. Referring to Example 48.17, find the following fixed fields in $E = \mathbb{Q}(\sqrt{2}, \sqrt{3})$.

a. $E_{\{\sigma_1,\sigma_3\}}$ **b.** $E_{\{\sigma_3\}}$ **c.** $E_{\{\sigma_2,\sigma_3\}}$

In Exercises 16 through 21, refer to the directions for Exercises 9 through 14 and find the fixed field of the automorphism or set of automorphisms of E.

16. τ_3

17. τ_3^2

18. $\{\tau_2, \tau_3\}$

19. $\tau_5\tau_2$

20. $\tau_5\tau_3\tau_2$

21. $\{\tau_2, \tau_3, \tau_5\}$

22. Refer to the directions for Exercises 9 through 14 for this exercise.

a. Show that each of the automorphisms τ_2, τ_3 and τ_5 is of order 2 in $G(E/\mathbb{Q})$. (Remember what is meant by the *order* of an element of a group.)

b. Find the subgroup H of $G(E/\mathbb{Q})$ generated by the elements τ_2, τ_3, and τ_5, and give the group table. [*Hint:* There are eight elements.]

c. Just as was done in Example 48.17, argue that the group H of part (b) is the full group $G(E/\mathbb{Q})$.

Concepts

In Exercises 23 and 24, correct the definition of the italicized term without reference to the text, if correction is needed, so that it is in a form acceptable for publication.

23. Two elements, α and β, of an algebraic extension E of a field F are *conjugate over* F if and only if they are both zeros of the same polynomial $f(x)$ in $F[x]$.

24. Two elements, α and β, of an algebraic extension E of a field F are *conjugate over* F if and only if the evaluation homomorphisms $\phi_\alpha : F[x] \to E$ and $\phi_\beta : F[x] \to E$ have the same kernel.

25. The fields $\mathbb{Q}(\sqrt{2})$ and $\mathbb{Q}(3+\sqrt{2})$ are the same, of course. Let $\alpha = 3 + \sqrt{2}$.

a. Find a conjugate $\beta \neq \alpha$ of α over $\mathbb{Q}$.

b. Referring to part (a), compare the conjugation automorphism $\psi_{\sqrt{2},-\sqrt{2}}$ of $\mathbb{Q}(\sqrt{2})$ with the conjugation automorphism $\psi_{\alpha,\beta}$.

26. Describe the value of the Frobenius automorphism σ_2 on each element of the finite field of four elements given in Example 29.19. Find the fixed field of σ_2.

27. Describe the value of the Frobenius automorphism σ_3 on each element of the finite field of nine elements given in Exercise 18 of Section 29. Find the fixed field of σ_3.

28. Let F be a field of characteristic $p \neq 0$. Give an example to show that the map $\sigma_p : F \to F$ given by $\sigma_p(a) = a^p$ for $a \in F$ need not be an automorphism in the case that F is infinite. What may go wrong?

29. Mark each of the following true or false.

_____ **a.** For all $\alpha, \beta \in E$, there is always an automorphism of E mapping α onto β.

_____ **b.** For α, β algebraic over a field F, there is always an isomorphism of $F(\alpha)$ onto $F(\beta)$.

_____ **c.** For α, β algebraic and conjugate over a field F, there is always an isomorphism of $F(\alpha)$ onto $F(\beta)$.

_____ **d.** Every automorphism of every field E leaves fixed every element of the prime subfield of E.

_____ **e.** Every automorphism of every field E leaves fixed an infinite number of elements of E.

_____ **f.** Every automorphism of every field E leaves fixed at least two elements of E.

_____ **g.** Every automorphism of every field E of characteristic 0 leaves fixed an infinite number of elements of E.

_____ **h.** All automorphisms of a field E form a group under function composition.

_____ **i.** The set of all elements of a field E left fixed by a single automorphism of E forms a subfield of E.

_____ **j.** For fields $F \leq E \leq K$, $G(K/E) \leq G(K/F)$.

Proof Synopsis

30. Give a one-sentence synopsis of the "if" part of Theorem 48.3.

31. Give a one-sentence synopsis of the "only if" part of Theorem 48.3.

Theory

32. Let α be algebraic of degree n over F. Show from Corollary 48.5 that there are at most n different isomorphisms of $F(\alpha)$ onto a subfield of $\overline{F}$ and leaving F fixed.

33. Let $F(\alpha_1, \cdots, \alpha_n)$ be an extension field of F. Show that any automorphism σ of $F(\alpha_1, \cdots, \alpha_n)$ leaving F fixed is completely determined by the n values $\sigma(\alpha_i)$.

34. Let E be an algebraic extension of a field F, and let σ be an automorphism of E leaving F fixed. Let $\alpha \in E$. Show that σ induces a permutation of the set of all zeros of $\text{irr}(\alpha, F)$ that are in E.

35. Let E be an algebraic extension of a field F. Let $S = \{\sigma_i \mid i \in I\}$ be a collection of automorphisms of E such that every σ_i leaves each element of F fixed. Show that if S generates the subgroup H of $G(E/F)$, then $E_S = E_H$.

36. We saw in Corollary 23.17 that the cyclotomic polynomial

$$\Phi_p(x) = \frac{x^p - 1}{x - 1} = x^{p-1} + x^{p-2} + \cdots + x + 1$$

is irreducible over $\mathbb{Q}$ for every prime p. Let ζ be a zero of $\Phi_p(x)$, and consider the field $\mathbb{Q}(\zeta)$.

a. Show that $\zeta, \zeta^2, \cdots, \zeta^{p-1}$ are distinct zeros of $\Phi_p(x)$, and conclude that they are all the zeros of $\Phi_p(x)$.

b. Deduce from Corollary 48.5 and part (a) of this exercise that $G(\mathbb{Q}(\zeta)/\mathbb{Q})$ is abelian of order $p - 1$.

c. Show that the fixed field of $G(\mathbb{Q}(\zeta)/\mathbb{Q})$ is $\mathbb{Q}$. [*Hint:* Show that

$$\{\zeta, \zeta^2, \cdots, \zeta^{p-1}\}$$

is a basis for $\mathbb{Q}(\zeta)$ over $\mathbb{Q}$, and consider which linear combinations of $\zeta, \zeta^2, \cdots, \zeta^{p-1}$ are left fixed by all elements of $G(\mathbb{Q}(\zeta)/\mathbb{Q})$.

37. Theorem 48.3 described conjugation isomorphisms for the case where α and β were conjugate algebraic elements over F. Is there a similar isomorphism of $F(\alpha)$ with $F(\beta)$ in the case that α and β are both transcendental over F?

38. Let F be a field, and let x be an indeterminate over F. Determine all automorphisms of $F(x)$ leaving F fixed, by describing their values on x.

39. Prove the following sequence of theorems.

a. An automorphism of a field E carries elements that are squares of elements in E onto elements that are squares of elements of E.

b. An automorphism of the field $\mathbb{R}$ of real numbers carries positive numbers onto positive numbers.

c. If σ is an automorphism of $\mathbb{R}$ and $a < b$, where $a, b \in \mathbb{R}$, then $\sigma(a) < \sigma(b)$.

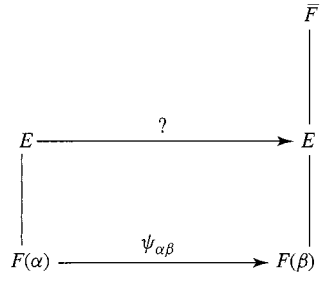
d. The only automorphism of $\mathbb{R}$ is the identity automorphism.

SECTION 49 THE ISOMORPHISM EXTENSION THEOREM

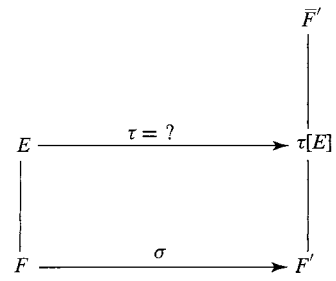
The Extension Theorem

Let us continue studying automorphisms of fields. In this section and the next, we shall be concerned with both the existence and the number of automorphisms of a field E.

Suppose that E is an algebraic extension of F and that we want to find some automorphisms of E. We know from Theorem 48.3 that if $\alpha, \beta \in E$ are conjugate over F, then there is an isomorphism $\psi_{\alpha,\beta}$ of $F(\alpha)$ onto $F(\beta)$. Of course, $\alpha, \beta \in E$ implies both $F(\alpha) \leq E$ and $F(\beta) \leq E$. It is natural to wonder whether the domain of definition of $\psi_{\alpha,\beta}$ can be enlarged from $F(\alpha)$ to a larger field, perhaps all of E, and whether this might perhaps lead to an automorphism of E. A mapping diagram of this situation is shown in Fig. 49.1. Rather than speak of "enlarging the domain of definition of $\psi_{\alpha,\beta}$," it



49.1 Figure



49.2 Figure

is customary to speak of "**extending the map $\psi_{\alpha,\beta}$ to a map τ**," which is a mapping of all of E.

Remember that we are always assuming that all algebraic extension of F under consideration are contained in a fixed algebraic closure $\overline{F}$ of F. The Isomorphism Extension Theorem shows that the mapping $\psi_{\alpha,\beta}$ can indeed always be extended to an *isomorphism* of E onto a subfield of $\overline{F}$. Whether this extension gives an *automorphism* of E, that is, maps E into itself, is a question we shall study in Section 50. Thus this extension theorem, used in conjunction with our conjugation isomorphisms $\psi_{\alpha,\beta}$ will guarantee the existence of lots of *isomorphism mappings*, at least, for many fields. Extension theorems are very important in mathematics, particularly in algebraic and topological situations.

Let us take a more general look at this situation. Suppose that E is an algebraic extension of a field F and that we have an isomorphism σ of F onto a field F'. Let $\overline{F'}$ be an algebraic closure of F'. We would like to extend σ to an isomorphism τ of E onto a subfield of $\overline{F'}$. This situation is shown in Fig. 49.2. Naively, we pick $\alpha \in E$ but not in F and try to extend σ to $F(\alpha)$. If

$$p(x) = \text{irr}(\alpha, F) = a_0 + a_1 x + \cdots + a_n x^n,$$

let β be a zero in $\overline{F'}$ of

$$q(x) = \sigma(a_0) + \sigma(a_1)x + \cdots + \sigma(a_n)x^n.$$

Here $q(x) \in F'[x]$. Since σ is an isomorphism, we know that $q(x)$ is irreducible in $F'[x]$. It seems reasonable that $F(\alpha)$ can be mapped isomorphically onto $F'(\beta)$ by a map extending σ and mapping α onto β. (This is not quite Theorem 48.3, but it is close to it; a few elements have been renamed by the isomorphism σ.) If $F(\alpha) = E$, we are done. If $F(\alpha) \neq E$, we have to find another element in E not in $F(\alpha)$ and continue the process. It is a situation very much like that in the construction of an algebraic closure $\overline{F}$ of a field F. Again the trouble is that, in general, where E is not a finite extension, the process may have to be repeated a (possibly large) infinite number of times, so we need Zorn's lemma to handle it. For this reason, we postpone the general proof of Theorem 49.3 to the end of this section.

49.3 Theorem **(Isomorphism Extension Theorem)** Let E be an algebraic extension of a field F. Let σ be an isomorphism of F onto a field F'. Let $\overline{F'}$ be an algebraic closure of F'. Then σ

can be extended to an isomorphism τ of E onto a subfield of $\overline{F'}$ such that $\tau(a) = \sigma(a)$ for all $a \in F$.

We give as a corollary the existence of an extension of one of our conjugation isomorphisms $\psi_{\alpha,\beta}$, as discussed at the start of this section.

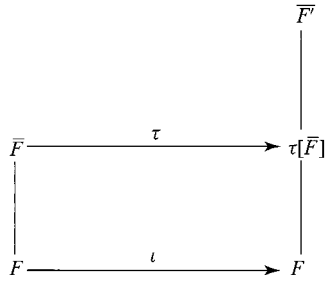
49.4 Corollary If $E \leq \overline{F}$ is an algebraic extension of F and $\alpha, \beta \in E$ are conjugate over F, then the conjugation isomorphism $\psi_{\alpha,\beta} : F(\alpha) \to F(\beta)$, given by Theorem 48.3, can be extended to an isomorphism of E onto a subfield of $\overline{F}$.

Proof Proof of this corollary is immediate from Theorem 49.3 if in the statement of the theorem we replace F by $F(\alpha)$, F' by $F(\beta)$, and $\overline{F'}$ by $\overline{F}$. ◆

As another corollary, we can show, as we promised earlier, that an algebraic closure of F is unique, up to an isomorphism leaving F fixed.

49.5 Corollary Let $\overline{F}$ and $\overline{F'}$ be two algebraic closures of F. Then $\overline{F}$ is isomorphic to $\overline{F'}$ under an isomorphism leaving each element of F fixed.

Proof By Theorem 49.3, the identity isomorphism of F onto F can be extended to an isomorphism τ mapping $\overline{F}$ onto a subfield of $\overline{F'}$ that leaves F fixed (see Fig. 49.6). We need only show that τ is onto $\overline{F'}$. But by Theorem 49.3, the map $\tau^{-1} : \tau[F] \to \overline{F}$ can be extended to an isomorphism of $\overline{F'}$ onto a subfield of $\overline{F}$. Since τ^{-1} is already onto $\overline{F}$, we must have $\tau[\overline{F}] = \overline{F'}$. ◆



49.6 Figure

The Index of a Field Extension

Having discussed the question of *existence*, we turn now to the question of *how many*. For a *finite* extension E of a field F, we would like to count how many isomorphisms there are of E onto a subfield of $\overline{F}$ that leave F fixed. We shall show that there are only a finite number of isomorphisms. Since every automorphism in $G(E/F)$ is such an isomorphism. a count of these isomorphisms will include all these automorphisms. Example 48.17 showed that $G(\mathbb{Q}(\sqrt{2}, \sqrt{3})/\mathbb{Q})$ has four elements, and that $4 = [\mathbb{Q}(\sqrt{2}, \sqrt{3}) : \mathbb{Q}]$. While such an equality is not always true, it is true in a very important case. The next theorem

takes the first big step in proving this. We state the theorem in more general terms than we shall need, but it does not make the proof any harder.

49.7 Theorem Let E be a finite extension of a field F. Let σ be an isomorphism of F onto a field F', and let $\overline{F'}$ be an algebraic closure of F'. Then the number of extensions of σ to an isomorphism τ of E onto a subfield of $\overline{F'}$ is finite, and independent of F', $\overline{F'}$, and σ. That is, the number of extensions is completely determined by the two fields E and F; it is intrinsic to them.

Proof The diagram in Fig. 49.8 may help us to follow the construction that we are about to make. This diagram is constructed in the following way. Consider two isomorphisms

$$\sigma_1 : F \xrightarrow{\text{onto}} F_1', \qquad \sigma_2 : F \xrightarrow{\text{onto}} F_2',$$

where $\overline{F_1'}$ and $\overline{F_2'}$ are algebraic closures of F_1' and F_2', respectively. Now $\sigma_2\sigma_1^{-1}$ is an isomorphism of F_1' onto F_2'. Then by Theorem 49.3 and Corollary 49.5 there is an isomorphism

$$\lambda : \overline{F_1'} \xrightarrow{\text{onto}} \overline{F_2'}$$

extending this isomorphism $\sigma_2\sigma_1^{-1} : F_1' \xrightarrow{\text{onto}} F_2'$. Referring to Fig. 49.8, corresponding to each $\tau_1 : E \to \overline{F_1'}$ that extends σ_1 we obtain an isomorphism $\tau_2 : E \to \overline{F_2'}$, by starting at E and going first to the left, then up, and then to the right. Written algebraically,

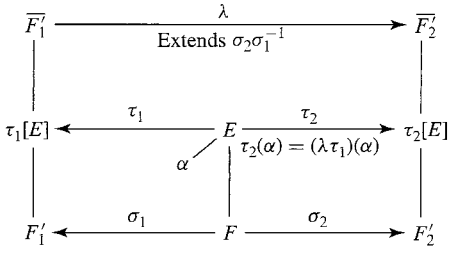
$$\tau_2(\alpha) = (\lambda\tau_1)(\alpha)$$

for $\alpha \in E$. Clearly τ_2 extends σ_2. The fact that we could have *started* with τ_2 and recovered τ_1 by defining

$$\tau_1(\alpha) = (\lambda^{-1}\tau_2)(\alpha),$$

that is, by chasing the other way around the diagram, shows that the correspondence between $\tau_1 : E \to \overline{F_1'}$ and $\tau_2 : E \to \overline{F_2'}$ is one to one. In view of this one-to-one correspondence, the number of τ extending σ is independent of F', $\overline{F'}$ and σ.

That the number of mappings extending σ is finite follows from the fact that since E is a finite extension of F, $E = F(\alpha_1, \cdots, \alpha_n)$ for some $\alpha_1, \cdots, \alpha_n$ in E, by Theorem 31.11.



49.8 Figure

There are only a finite number of possible candiates for the images $\tau(\alpha_i)$ in F', for if

$$\text{irr}(\alpha_i, F) = a_{i0} + a_{i1}x + \cdots + a_{im_i}x^{m_i},$$

where $a_{ik} \in F$, then $\tau(\alpha_i)$ must be one of the zeros in $\overline{F'}$ of

$$[\sigma(a_{i0}) + \sigma(a_{i1})x + \cdots + \sigma(a_{im_i})x^{m_i}] \in F'[x].$$

◆

49.9 Definition Let E be a finite extension of a field F. The number of isomorphisms of E onto a subfield of $\overline{F}$ leaving F fixed is the **index** $\{E : F\}$ **of** E **over** F. ■

49.10 Corollary If $F \leq E \leq K$, where K is a finite extension field of the field F, then $\{K : F\} = \{K : E\}\{E : F\}$.

Proof It follows from Theorem 49.7 that each of the $\{E : F\}$ isomorphisms τ_i of E onto a subfield of $\overline{F}$ leaving F fixed has $\{K : E\}$ extensions to an isomorphism of K onto a subfield of $\overline{F}$. ◆

The preceding corollary was really the main thing we were after. Note that it counts something. *Never underestimate a result that counts something*, even if it is only called a "corollary."

We shall show in Section 51 that unless F is an infinite field of characteristics $p \neq 0$, we always have $[E : F] = \{E : F\}$ for every finite extension field E of F. For the case $E = F(\alpha)$, the $\{F(\alpha) : F\}$ extensions of the identity map $\iota : F \to F$ to maps of $F(\alpha)$ onto a subfield of $\overline{F}$ are given by the conjugation isomorphisms $\psi_{\alpha,\beta}$ for each conjugate β in $\overline{F}$ of α over F. Thus if $\text{irr}(\alpha, F)$ has n *distinct* zeros in $\overline{F}$, we have $\{E : F\} = n$. We shall show later that unless F is infinite and of characteristic $p \neq 0$, the number of distinct zeros of $\text{irr}(\alpha, F)$ is $\deg(\alpha, F) = [F(\alpha) : F]$.

49.11 Example Consider $E = \mathbb{Q}(\sqrt{2}, \sqrt{3})$ over $\mathbb{Q}$, as in Example 48.17. Our work in that example shows that $\{E : \mathbb{Q}\} = [E : \mathbb{Q}] = 4$. Also, $\{E : \mathbb{Q}(\sqrt{2})\} = 2$, and $\{\mathbb{Q}(\sqrt{2}) : \mathbb{Q}\} = 2$, so

$$4 = \{E : \mathbb{Q}\} = \{E : \mathbb{Q}(\sqrt{2})\}\{\mathbb{Q}(\sqrt{2}) : \mathbb{Q}\} = (2)(2).$$

This illustrates Corollary 49.10 ▲

Proof of the Extension Theorem

We restate the Isomorphism Extension Theorem 49.3.

Isomorphism Extension Theorem Let E be an algebraic extension of a field F. Let σ be an isomorphism of F onto a field F'. Let $\overline{F'}$ be an algebraic closure of F'. Then σ can be extended to an isomorphism τ of E onto a subfield of $\overline{F'}$ such that $\tau(a) = \sigma(a)$ for $a \in F$.

Proof Consider all pairs (L, λ), where L is a field such that $F \leq L \leq E$ and λ is an isomorphism of L onto a subfield of $\overline{F'}$ such that $\lambda(a) = \sigma(a)$ for $a \in F$. The set S of such

pairs (L, λ) is nonempty, since (F, σ) is such a pair. Define a partial ordering on S by $(L_1, \lambda_1) \leq (L_2, \lambda_2)$, if $L_1 \leq L_2$ and $\lambda_1(a) = \lambda_2(a)$ for $a \in L_1$. It is readily checked that this relation $\leq$ does give a partial ordering of S.

Let $T = \{(H_i, \lambda_i) \mid i \in I\}$ be a chain of S. We claim that $H = \bigcup_{i \in I} H_i$ is a subfield of E. Let $a, b \in H$, where $a \in H_1$ and $b \in H_2$; then either $H_1 \leq H_2$ or $H_2 \leq H_1$, since T is a chain. If, say, $H_1 \leq H_2$, then $a, b, \in H_2$, so $a \pm b$, ab, and a/b for $b \neq 0$ are all in H_2 and hence in H. Since for each $i \in I$, $F \subseteq H_i \subseteq E$, we have $F \subseteq H \subseteq E$. Thus H is a subfield of E.

Define $\lambda : H \to \overline{F'}$ as follows. Let $c \in H$. Then $c \in H_i$ for some $i \in I$, and let

$$\lambda(c) = \lambda_i(c).$$

The map λ is well defined because if $c \in H_1$ and $c \in H_2$, then either $(H_1, \lambda_1) \leq (H_2, \lambda_2)$ or $(H_2, \lambda_2) \leq (H_1, \lambda_1)$, since T is a chain. In either case, $\lambda_1(c) = \lambda_2(c)$. We claim that λ is an isomorphism of H onto a subfield of $\overline{F'}$. If $a, b \in H$ then there is an H_i such that $a, b \in H_i$, and

$$\lambda(a + b) = \lambda_i(a + b) = \lambda_i(a) + \lambda_i(b) = \lambda(a) + \lambda(b).$$

Similarly,

$$\lambda(ab) = \lambda_i(ab) = \lambda_i(a)\lambda_i(b) = \lambda(a)\lambda(b).$$

If $\lambda(a) = 0$, then $a \in H_i$ for some i implies that $\lambda_i(a) = 0$, so $a = 0$. Therefore, λ is an isomorphism. Thus $(H, \lambda) \in S$, and it is clear from our definitions of H and λ that (H, λ) is an upper bound for T.

We have shown that every chain of S has an upper bound in S, so the hypotheses of Zorn's lemma are satisfied. Hence there exists a maximal element (K, τ) of S. Let $\tau(K) = K'$, where $K' \leq \overline{F'}$. Now if $K \neq E$, let $\alpha \in E$ but $\alpha \notin K$. Now α is algebraic over F, so α is algebraic over K. Also, let $p(x) = \text{irr}(\alpha, K)$. Let ψ_α be the canonical isomorphism

$$\psi_\alpha : K[x]/\langle p(x)\rangle \to K(\alpha),$$

corresponding to the evaluation homomorphism $\phi_\alpha : K[x] \to K(\alpha)$. If

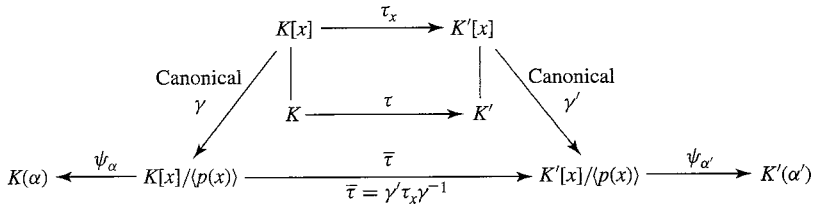
$$p(x) = a_0 + a_1 x + \cdots + a_n x^n,$$

consider

$$q(x) = \tau(a_0) + \tau(a_1)x + \cdots + \tau(a_n)x^n$$

in $K'[x]$. Since τ is an isomorphism, $q(x)$ is irreducible in $K'[x]$. Since $K' \leq F'$, there is a zero α' of $q(x)$ in $\overline{F'}$. Let

$$\psi_{\alpha'} : K'[x]/\langle q(x)\rangle \to K'(\alpha')$$



49.12 Figure

be the isomorphism analogous to ψ_α. Finally, let

$$\bar{\tau} : K[x]/\langle p(x)\rangle \to K'[x]/\langle q(x)\rangle$$

be the isomorphism extending τ on K and mapping $x + \langle p(x)\rangle$ onto $x + \langle q(x)\rangle$. (See Fig. 49.12.) Then the composition of maps

$$\psi_{\alpha'}\bar{\tau}\psi_\alpha^{-1} : K(\alpha) \to K'(\alpha')$$

is an isomorphism of $K(\alpha)$ onto a subfield of $\overline{F'}$. Clearly, $(K, \tau) < (K(\alpha), \psi_{\alpha'}\bar{\tau}\psi_\alpha^{-1})$, which contradicts that (K, τ) is maximal. Therefore we must have had $K = E$. ◆

■ EXERCISES 49

Computations

Let $E = \mathbb{Q}(\sqrt{2}, \sqrt{3}, \sqrt{5})$. It can be shown that $[E : \mathbb{Q}] = 8$. In Exercises 1 through 3, for the given isomorphic mapping of a subfield of E, give all extensions of the mapping to an isomorphic mapping of E onto a subfield of $\overline{\mathbb{Q}}$. Describe the extensions by giving values on the generating set $\{\sqrt{2}, \sqrt{3}, \sqrt{5}\}$ for E over $\mathbb{Q}$.

1. $\iota : \mathbb{Q}(\sqrt{2}, \sqrt{15}) \to \mathbb{Q}(\sqrt{2}, \sqrt{15})$, where ι is the identity map
2. $\sigma : \mathbb{Q}(\sqrt{2}, \sqrt{15}) \to \mathbb{Q}(\sqrt{2}, \sqrt{15})$ where $\sigma(\sqrt{2}) = \sqrt{2}$ and $\sigma(\sqrt{15}) = -\sqrt{15}$
3. $\psi_{\sqrt{30},-\sqrt{30}} : \mathbb{Q}(\sqrt{30}) \to \mathbb{Q}(\sqrt{30})$

It is a fact, which we can verify by cubing, that the zeros of $x^3 - 2$ in $\mathbb{Q}$ are

$$\alpha_1 = \sqrt[3]{2}, \qquad \alpha_2 = \sqrt[3]{2}\frac{-1 + i\sqrt{3}}{2}, \qquad \text{and} \qquad \alpha_3 = \sqrt[3]{2}\frac{-1 - i\sqrt{3}}{2},$$

where $\sqrt[3]{2}$, as usual, is the real cube root of 2. Use this information in Exercises 4 through 6.

4. Describe all extensions of the identity map of $\mathbb{Q}$ to an isomorphism mapping $\mathbb{Q}(\sqrt[3]{2})$ onto a subfield of $\overline{\mathbb{Q}}$.
5. Describe all extensions of the identity map of $\mathbb{Q}$ to an isomorphism mapping $\mathbb{Q}(\sqrt[3]{2}, \sqrt{3})$ onto a subfield of $\overline{\mathbb{Q}}$.
6. Describe all extensions of the automorphism $\psi_{\sqrt{3},-\sqrt{3}}$ of $\mathbb{Q}(\sqrt{3})$ to an isomorphism mapping $\mathbb{Q}(i, \sqrt{3}, \sqrt[3]{2})$ onto a subfield of $\overline{\mathbb{Q}}$.
7. Let σ be the automorphism of $\mathbb{Q}(\pi)$ that maps π onto $-\pi$.
   a. Describe the fixed field of σ.
   b. Describe all extensions of σ to an isomorphism mapping the field $\mathbb{Q}(\sqrt{\pi})$ onto a subfield of $\overline{\mathbb{Q}(\pi)}$.

Concepts

8. Mark each of the following true or false.

_____ **a.** Let $F(\alpha)$ be any simple extension of a field F. Then every isomorphism of F onto a subfield of $\bar{F}$ has an extension to an isomorphism of $F(\alpha)$ onto a subfield of $\bar{F}$.

_____ **b.** Let $F(\alpha)$ be any simple algebraic extension of a field F. Then every isomorphism of F onto a subfield of $\bar{F}$ has an extension to an isomorphism of $F(\alpha)$ onto a subfield of $\bar{F}$.

_____ **c.** An isomorphism of F onto a subfield of $\bar{F}$ has the same number of extensions to each simple algebraic extension of F.

_____ **d.** Algebraic closures of isomorphic fields are always isomorphic.

_____ **e.** Algebraic closures of fields that are not isomorphic are never isomorphic.

_____ **f.** Any algebraic closure of $\mathbb{Q}(\sqrt{2})$ is isomorphic to any algebraic closure of $\mathbb{Q}(\sqrt{17})$.

_____ **g.** The index of a finite extension E over a field F is finite.

_____ **h.** The index behaves multiplicatively with respect to finite towers of finite extensions of fields.

_____ **i.** Our remarks prior to the first statement of Theorem 49.3 essentially constitute a proof of this theorem for a finite extension E over F.

_____ **j.** Corollary 49.5 shows that $\mathbb{C}$ is isomorphic to $\overline{\mathbb{Q}}$.

Theory

9. Let K be an algebraically closed field. Show that every isomorphism σ of K onto a subfield of itself such that K is algebraic over $\sigma[K]$ is an automorphism of K, that is, is an onto map. [*Hint:* Apply Theorem 49.3 to σ^{-1}.]

10. Let E be an algebraic extension of a field F. Show that every isomorphism of E onto a subfield of $\bar{F}$ leaving F fixed can be extended to an automorphism of $\bar{F}$.

11. Prove that if E is an algebraic extension of a field F, then two algebraic closures $\bar{F}$ and $\overline{E}$ of F and E, respectively, are isomorphic.

12. Prove that the algebraic closure of $\mathbb{Q}(\sqrt{\pi})$ in $\mathbb{C}$ is isomorphic to any algebraic closure of $\overline{\mathbb{Q}}(x)$, where $\overline{\mathbb{Q}}$ is the field of algebraic numbers and x is an indeterminate.

13. Prove that if E is a finite extension of a field F, then $\{E : F\} \leq [E : F]$. [*Hint:* The remarks preceding Example 49.11 essentially showed this for a simple algebraic extension $F(\alpha)$ of F. Use the fact that a finite extension is a tower of simple extensions, together with the multiplicative properties of the index and degree.]

SECTION 50 Splitting Fields

We are going to be interested chiefly in *automorphisms* of a field E, rather than mere isomorphic mappings of E onto a subfield of $\bar{E}$. It is the *automorphisms* of a field that form a group. We wonder whether for some extension field E of a field F, *every* isomorphic mapping of E onto a subfield of $\bar{F}$ leaving F fixed is actually an automorphism of E.

Suppose E is an algebraic extension of a field E. If $\alpha \in E$ and $\beta \in \bar{F}$ is a conjugate of α over F, then there is a conjugation isomorphism

$$\psi_{\alpha,\beta} : F(\alpha) \to F(\beta).$$

By Corollary 49.4, $\psi_{\alpha,\beta}$ can be extended to an isomorphic mapping of E onto a subfield of $\bar{F}$. Now if $\beta \notin E$, such an isomorphic mapping of E can't be an automorphism of E. *Thus, if an algebraic extension E of a field F is such that all its isomorphic mappings onto*

a subfield of $\bar{F}$ leaving F fixed are actually automorphisms of E, then for every $\alpha \in E$, all conjugates of α over F must be in E also. This observation seemed to come very easily. We point out that we used a lot of power, namely the existence of the conjugation isomorphisms and the Isomorphism Extension Theorem 49.3.

These ideas suggest the formulation of the following definition.

50.1 Definition Let F be a field with algebraic closure $\bar{F}$. Let $\{f_i(x) \mid i \in I\}$ be a collection of polynomials in $F[x]$. A field $E \leq \bar{F}$ is the **splitting field of** $\{f_i(x) \mid i \in I\}$ **over** F if E is the smallest subfield of $\bar{F}$ containing F and all the zeros in $\bar{F}$ of each of the $f_i(x)$ for $i \in I$. A field $K \leq \bar{F}$ is a **splitting field over** F if it is the splitting field of some set of polynomials in $F[x]$. ■

50.2 Example We see that $\mathbb{Q}[\sqrt{2}, \sqrt{3}]$ is a splitting field of $\{x^2 - 2, x^2 - 3\}$ and also of $\{x^4 - 5x^2 + 6\}$. ▲

For one polynomial $f(x) \in F[x]$, we shall often refer to the splitting field of $\{f(x)\}$ over F as the **splitting field of** $f(x)$ **over** F. Note that the splitting field of $\{f_i(x) \mid i \in I\}$ over F in $\bar{F}$ is the intersection of all subfields of $\bar{F}$ containing F and all zeros in $\bar{F}$ of each $f_i(x)$ for $i \in I$. Thus such a splitting field surely does exist.

We now show that splitting fields over F are precisely those fields $E \leq \bar{F}$ with the property that all isomorphic mappings of E onto a subfield of $\bar{F}$ leaving F fixed are automorphisms of E. This will be a corollary of the next theorem. *Once more, we are characterizing a concept in terms of mappings.* Remember, we are always assuming that all algebraic extensions of a field F under consideration are in one fixed algebraic closure $\bar{F}$ of F.

50.3 Theorem A field E, where $F \leq E \leq \bar{F}$, is a splitting field over F if and only if every automorphism of $\bar{F}$ leaving F fixed maps E onto itself and thus induces an automorphism of E leaving F fixed.

Proof Let E be a splitting field over F in $\bar{F}$ of $\{f_i(x) \mid i \in I\}$, and let σ be an automorphism of $\bar{F}$ leaving F fixed. Let $\{\alpha_j \mid j \in J\}$ be the collection of all zeros in $\bar{F}$ of all the $f_i(x)$ for $i \in I$. Now our previous work shows that for a fixed α_j, the field $F(\alpha_j)$ has as elements all expressions of the form

$$g(\alpha_j) = a_0 + a_1\alpha_j + \cdots + a_{n_j-1}\alpha_j^{n_j-1},$$

where n_j is the degree of $\text{irr}(\alpha_j, F)$ and $a_k \in F$. Consider the set S of all *finite* sums of *finite* products of elements of the form $g(\alpha_j)$ for all $j \in J$. The set S is a subset of E closed under addition and multiplication and containing 0, 1, and the additive inverse of each element. Since each element of S is in some $F(\alpha_{j_1}, \cdots, \alpha_{j_r}) \subseteq S$, we see that S also contains the multiplicative inverse of each nonzero element. Thus S is a subfield of E containing all α_j for $j \in J$. By definition of the splitting field E of $\{f_i(x) \mid i \in I\}$, we see that we must have $S = E$. All this work was just to show that $\{\alpha_j \mid j \in J\}$ *generates* E over F, in the sense of taking *finite* sums and *finite* products. Knowing this, we see immediately that the value of σ on any element of E is completely determined by the values $\sigma(\alpha_j)$. But by Corollary 48.5, $\sigma(\alpha_j)$ must also be a zero of $\text{irr}(\alpha_j, F)$. By

Theorem 29.13, $\text{irr}(\alpha_j, F)$ divides the $f_i(x)$ for which $f_i(\alpha_j) = 0$, so $\sigma(\alpha_j) \in E$ also. Thus σ maps E onto a subfield of E isomorphically. However, the same is true of the automorphism σ^{-1} of $\bar{F}$. Since for $B \in E$,

$$\beta = \sigma(\sigma^{-1}(\beta)),$$

we see that σ maps E onto E, and thus induces an automorphism of E.

Suppose, conversely, that every automorphism of $\bar{F}$ leaving F fixed induces an automorphism of E. Let $g(x)$ be an *irreducible* polynomial in $F[x]$ having a zero α in E. If β is any zero of $g(x)$ in $\bar{F}$, then by Theorem 48.3, there is a conjugation isomorphism $\psi_{\alpha,\beta}$ of $F(\alpha)$ onto $F(\beta)$ leaving F fixed. By Theorem 49.3, $\psi_{\alpha,\beta}$ can be extended to an isomorphism τ of $\bar{F}$ onto a subfield of $\bar{F}$. But then

$$\tau^{-1} : \tau[\bar{F}] \to \bar{F}$$

can be extended to an isomorphism mapping $\bar{F}$ onto a subfield of $\bar{F}$. Since the image of τ^{-1} is already all of $\bar{F}$, we see that τ must have been onto $\bar{F}$, so τ is an automorphism of $\bar{F}$ leaving F fixed. Then by assumption, τ induces an automorphism of E, so $\tau(\alpha) = \beta$ is in E. We have shown that if $g(x)$ is an irreducible polynomial in $F[x]$ having one zero in E, then all zeros of $g(x)$ in $\bar{F}$ are in E. Hence if $\{g_k(x)\}$ is the set of *all* irreducible polynomials in $F[x]$ having a zero in E, then E is the splitting field of $\{g_k(x)\}$. ◆

50.4 Definition Let E be an extension field of a field F. A polynomial $f(x) \in F[x]$ **splits in** E if it factors into a product of linear factors in $E[x]$. ■

50.5 Example The polynomial $x^4 - 5x^2 + 6$ in $\mathbb{Q}[x]$ splits in the field $\mathbb{Q}[\sqrt{2}, \sqrt{3}]$ into $(x - \sqrt{2})(x + \sqrt{2})(x - \sqrt{3})(x + \sqrt{3})$. ▲

50.6 Corollary If $E \leq \bar{F}$ is a splitting field over F, then every irreducible polynomial in $F[x]$ having a zero in E splits in E.

Proof If E is a splitting field over F in $\bar{F}$, then every automorphism of $\bar{F}$ induces an automorphism of E. The second half of the proof of Theorem 50.3 showed precisely that E is also the splitting field over F of the set $\{g_k(x)\}$ of *all* irreducible polynomials in $F[x]$ having a zero in E. Thus an irreducible polynomial $f(x)$ of $F[x]$ having a zero in E has all its zeros in $\bar{F}$ in E. Therefore, its factorization into linear factors in $\bar{F}[x]$, given by Theorem 31.15, actually takes place in $E[x]$, so $f(x)$ splits in E. ◆

50.7 Corollary If $E \leq \bar{F}$ is a splitting field over F, then every isomorphic mapping of E onto a subfield of $\bar{F}$ and leaving F fixed is actually an automorphism of E. In particular, if E is a splitting field of finite degree over F, then

$$\{E : F\} = |G(E/F)|.$$

Proof Every isomorphism σ mapping E onto a subfield of $\bar{F}$ leaving F fixed can be extended to an automorphism τ of $\bar{F}$, by Theorem 49.3, together with the *onto* argument of the second half of the proof of Theorem 50.3. If E is a splitting field over F, then by Theorem 50.3, τ restricted to E, that is σ, is an automorphism of E. Thus for a splitting field E over F, every isomorphic mapping of E onto a subfield of $\bar{F}$ leaving F fixed is an automorphism of E.

The equation $\{E : F\} = |G(E/F)|$ then follows immediately for a splitting field E of finite degree over F, since $\{E : F\}$ was defined as the number of different isomorphic mappings of E onto a subfield of $\overline{F}$ leaving F fixed. ◆

50.8 Example Observe that $\mathbb{Q}(\sqrt{2}, \sqrt{3})$ is the splitting field of

$$\{x^2 - 2, x^2 - 3\}$$

over $\mathbb{Q}$. Example 48.17 showed that the mappings ι, σ_1, σ_2, and σ_3 are all the automorphisms of $\mathbb{Q}(\sqrt{2}, \sqrt{3})$ leaving $\mathbb{Q}$ fixed. (Actually, since every automorphism of a field must leave the prime subfield fixed, we see that these are the only automorphisms of $\mathbb{Q}(\sqrt{2}, \sqrt{3})$.) Then

$$\{\mathbb{Q}(\sqrt{2}, \sqrt{3}) : \mathbb{Q}\} = |G(\mathbb{Q}(\sqrt{2}, \sqrt{3})/\mathbb{Q})| = 4.$$

illustrating Corollary 50.7. ▲

We wish to determine conditions under which

$$|G(E/F)| = \{E : F\} = [E : F]$$

for finite extensions E of F. This is our next topic. We shall show in the following section that this equation always holds when E is a splitting field over a field F of characteristic 0 or when F is a finite field. This equation need not be true when F is an infinite field of characteristic $p \neq 0$.

50.9 Example Let $\sqrt[3]{2}$ be the real cube root of 2, as usual. Now $x^3 - 2$ does not split in $\mathbb{Q}(\sqrt[3]{2})$, for $\mathbb{Q}(\sqrt[3]{2}) < \mathbb{R}$ and only one zero of $x^3 - 2$ is real. Thus $x^3 - 2$ factors in $(\mathbb{Q}(\sqrt[3]{2}))[x]$ into a linear factor $x - \sqrt[3]{2}$ and an irreducible quadratic factor. The splitting field E of $x^3 - 2$ over $\mathbb{Q}$ is therefore of degree 2 over $\mathbb{Q}(\sqrt[3]{2})$. Then

$$[E : \mathbb{Q}] = [E : \mathbb{Q}(\sqrt[3]{2})][\mathbb{Q}(\sqrt[3]{2}) : \mathbb{Q}] = (2)(3) = 6.$$

We have shown that the splitting field over $\mathbb{Q}$ of $x^3 - 2$ is of degree 6 over $\mathbb{Q}$.

We can verify by cubing that

$$\sqrt[3]{2}\frac{-1 + i\sqrt{3}}{2} \quad \text{and} \quad \sqrt[3]{2}\frac{-1 - i\sqrt{3}}{2}$$

are the other zeros of $x^3 - 2$ in $\mathbb{C}$. Thus the splitting field E of $x^3 - 2$ over $\mathbb{Q}$ is $\mathbb{Q}(\sqrt[3]{2}, i\sqrt{3})$. (This is *not* the same field as $\mathbb{Q}(\sqrt[3]{2}, i, \sqrt{3})$, which is of degree 12 over $\mathbb{Q}$.) Further study of this interesting example is left to the exercises (see Exercises 7, 8, 9, 16, 21, and 23). ▲

EXERCISES 50

Computations

In Exercises 1 through 6, find the degree over $\mathbb{Q}$ of the splitting field over $\mathbb{Q}$ of the given polynomial in $\mathbb{Q}[x]$.

1. $x^2 + 3$ **2.** $x^4 - 1$ **3.** $(x^2 - 2)(x^2 - 3)$

4. $x^3 - 3$ **5.** $x^3 - 1$ **6.** $(x^2 - 2)(x^3 - 2)$

Refer to Example 50.9 for Exercises 7 through 9.

7. What is the order of $G(\mathbb{Q}(\sqrt[3]{2})/\mathbb{Q})$?
8. What is the order of $G(\mathbb{Q}(\sqrt[3]{2}, i\sqrt{3})/\mathbb{Q})$?
9. What is the order of $G(\mathbb{Q}(\sqrt[3]{2}, i\sqrt{3})/\mathbb{Q}(\sqrt[3]{2}))$?
10. Let α be a zero of $x^3 + x^2 + 1$ over $\mathbb{Z}_2$. Show that $x^3 + x^2 + 1$ splits in $\mathbb{Z}_2(\alpha)$. [*Hint:* There are eight elements in $\mathbb{Z}_2(\alpha)$. Exhibit two more zeros of $x^3 + x^2 + 1$, in addition to α, among these eight elements. Alternatively, use the results of Section 33.]

Concepts

In Exercises 11 and 12, correct the definition of the italicized term without reference to the text, if correction is needed, so that it is in a form acceptable for publication.

11. Let $F \leq E \leq \bar{F}$ where $\bar{F}$ is an algebraic closure of a field F. The field E is a *splitting field* over F if and only if E contains all the zeros in $\bar{F}$ of every polynomial in $F[x]$ that has a zero in E.
12. A polynomial $f(x)$ in $F[x]$ *splits in an extension field* E of F if and only if it factors in $E[x]$ into a product of polynomials of lower degree.
13. Let $f(x)$ be a polynomial in $F[x]$ of degree n. Let $E \leq \bar{F}$ be the splitting field of $f(x)$ over F in $\bar{F}$. What bounds can be put on $[E : F]$?
14. Mark each of the following true or false.

    _____ **a.** Let $\alpha, \beta \in E$, where $E \leq \bar{F}$ is a splitting field over F. Then there exists an automorphism of E leaving F fixed and mapping α onto β if and only if $\text{irr}(\alpha, F) = \text{irr}(\beta, F)$.

    _____ **b.** $\mathbb{R}$ is a splitting field over $\mathbb{Q}$.

    _____ **c.** $\mathbb{R}$ is a splitting field over $\mathbb{R}$.

    _____ **d.** $\mathbb{C}$ is a splitting field over $\mathbb{R}$.

    _____ **e.** $\mathbb{Q}(i)$ is a splitting field over $\mathbb{Q}$.

    _____ **f.** $\mathbb{Q}(\pi)$ is a splitting field over $\mathbb{Q}(\pi^2)$.

    _____ **g.** For every splitting field E over F, where $E \leq \bar{F}$, every isomorphic mapping of E is an automorphism of E.

    _____ **h.** For every splitting field E over F, where $E \leq \bar{F}$, every isomorphism mapping E onto a subfield of $\bar{F}$ is an automorphism of E.

    _____ **i.** For every splitting field E over F, where $E \leq \bar{F}$, every isomorphism mapping E onto a subfield of $\bar{F}$ and leaving F fixed is an automorphism of E.

    _____ **j.** Every algebraic closure $\bar{F}$ of a field F is a splitting field over F.

15. Show by an example that Corollary 50.6 is no longer true if the word *irreducible* is deleted.
16. **a.** Is $|G(E/F)|$ multiplicative for finite towers of finite extensions, that is, is

$$|G(E/F)| = |G(K/E)||G(E/F)| \qquad \text{for} \qquad F \leq E \leq K \leq \bar{F}?$$

    Why or why not? [*Hint:* Use Exercises 7 through 9.]

    b. Is $|G(E/F)|$ multiplicative for finite towers of finite extensions, each of which is a splitting field over the bottom field? Why or why not?

Theory

17. Show that if a finite extension E of a field F is a splitting field over F, then E is a splitting field of one polynomial in $F[x]$.

18. Show that if $[E : F] = 2$, then E is a splitting field over F.

19. Show that for $F \leq E \leq \bar{F}$, E is a splitting field over F if and only if E contains all conjugates over F in $\bar{F}$ for each of its elements.

20. Show that $\mathbb{Q}(\sqrt[3]{2})$ has only the identity automorphism.

21. Referring to Example 50.9, show that

$$G(\mathbb{Q}(\sqrt[3]{2}, i\sqrt{3})/\mathbb{Q}(i\sqrt{3})) \simeq \langle \mathbb{Z}_3, + \rangle.$$

22. a. Show that an automorphism of a splitting field E over F of a polynomial $f(x) \in F[x]$ permutes the zeros of $f(x)$ in E.

b. Show that an automorphism of a splitting field E over F of a polynomial $f(x) \in F[x]$ is completely determined by the permutation of the zeros of $f(x)$ in E given in part (a).

c. Show that if E is a splitting field over F of a polynomial $f(x) \in F[x]$, then $G(E/F)$ can be viewed in a natural way as a certain group of permutations.

23. Let E be the splitting field of $x^3 - 2$ over $\mathbb{Q}$, as in Example 50.9.

a. What is the order of $G(E/\mathbb{Q})$? [*Hint:* Use Corollary 50.7 and Corollary 49.4 applied to the tower $\mathbb{Q} \leq \mathbb{Q}(i\sqrt{3}) \leq E$.]

b. Show that $G(E/\mathbb{Q}) = S_3$, the symmetric group on three letters. [*Hint:* Use Exercise 22, together with part (a).]

24. Show that for a prime p, the splitting field over $\mathbb{Q}$ of $x^p - 1$ is of degree $p - 1$ over $\mathbb{Q}$. [*Hint:* Refer to Corollary 23.17.]

25. Let $\bar{F}$ and $\bar{F}'$ be two algebraic closures of a field F, and let $f(x) \in F[x]$. Show that the splitting field E over F of $f(x)$ in $\bar{F}$ is isomorphic to the splitting field E' over F of $f(x)$ in $\bar{F}'$. [*Hint:* Use Corollary 49.5.]

SECTION 51 SEPARABLE EXTENSIONS

Multiplicity of Zeros of a Polynomial

Remember that we are now always assuming that all algebraic extensions of a field F under consideration are contained in one fixed algebraic closure $\bar{F}$ of F.

Our next aim is to determine, for a finite extension E of F, under what conditions $\{E : F\} = [E : F]$. The key to answering this question is to consider the multiplicity of zeros of polynomials.

51.1 Definition Let $f(x) \in F[x]$. An element α of $\bar{F}$ such that $f(\alpha) = 0$ is a **zero of $f(x)$ of multiplicity** ν if ν is the greatest integer such that $(x - \alpha)^\nu$ is a factor of $f(x)$ in $\bar{F}[x]$. ■

The next theorem shows that the multiplicities of the zeros of one given *irreducible* polynomial over a field are all the same. The ease with which we can prove this theorem is a further indication of the power of our conjugation isomorphisms and of our whole approach to the study of zeros of polynomials by means of mappings.

51.2 Theorem Let $f(x)$ be irreducible in $F[x]$. Then all zeros of $f(x)$ in $\bar{F}$ have the same multiplicity.

Proof Let α and β be zeros of $f(x)$ in $\bar{F}$. Then by Theorem 48.3, there is a conjugation isomorphism $\psi_{\alpha,\beta} : F(\alpha) \xrightarrow{\text{onto}} F(\beta)$. By Corollary 49.4, $\psi_{\alpha,\beta}$ can be extended to an isomorphism $\tau : \bar{F} \to \bar{F}$. Then τ induces a natural isomorphism $\tau_x : \bar{F}[x] \to \bar{F}[x]$, with $\tau_x(x) = x$. Now τ_x leaves $f(x)$ fixed, since $f(x) \in F[x]$ and $\psi_{\alpha,\beta}$ leaves F fixed. However,

$$\tau_x((x - \alpha)^\nu) = (x - \beta)^\nu,$$

which shows that the multiplicity of β in $f(x)$ is greater than or equal to the multiplicity of α. A symmetric argument gives the reverse inequality, so the multiplicity of α equals that of β. ◆

51.3 Corollary If $f(x)$ is irreducible in $F[x]$, then $f(x)$ has a factorization in $F[x]$ of the form

$$a \prod_i (x - \alpha_i)^\nu,$$

where the α_i are the distinct zeros of $f(x)$ in $\bar{F}$ and $a \in F$.

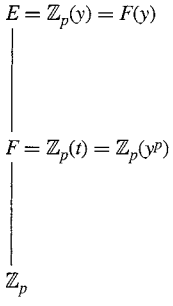
Proof The corollary is immediate from Theorem 51.2. ◆

At this point, we should probably show by an example that the phenomenon of a zero of multiplicity greater than 1 of an irreducible polynomial can occur. We shall show later in this section that it can only occur for a polynomial over an infinite field of characteristic $p \neq 0$.

51.4 Example Let $E = \mathbb{Z}_p(y)$, where y is an indeterminate. Let $t = y^p$, and let F be the subfield $\mathbb{Z}_p(t)$ of E. (See Fig. 51.5.) Now $E = F(y)$ is algebraic over F, for y is a zero of $(x^p - t) \in F[x]$. By Theorem 29.13, $\text{irr}(y, F)$ must divide $x^p - t$ in $F[x]$. [Actually, $\text{irr}(y, F) = x^p - t$. We leave a proof of this to the exercises (see Exercise 10).] Since $F(y)$ is not equal to F, we must have the degree of $\text{irr}(y, F) \geq 2$. But note that

$$x^p - t = x^p - y^p = (x - y)^p,$$

since E has characteristic p (see Theorem 48.19 and the following comment). Thus y is a zero of $\text{irr}(y, F)$ of multiplicity > 1. Actually, $x^p - t = \text{irr}(y, F)$, so the multiplicity of y is p. ▲



51.5 Figure

From here on we rely heavily on Theorem 49.7 and its corollary. Theorem 48.3 and its corollary show that for a simple algebraic extension $F(\alpha)$ of F there is one extension of the identity isomorphism ι mapping F into F for every distinct zero of $\text{irr}(\alpha, F)$ and that these are the only extensions of ι. *Thus $\{F(\alpha) : F\}$ is the number of distinct zeros of* $\text{irr}(\alpha, F)$.

In view of our work with the theorem of Lagrange and Theorem 31.4, we should recognize the potential of a theorem like this next one.

51.6 Theorem If E is a finite extension of F, then $\{E : F\}$ divides $[E : F]$.

Proof By Theorem 31.11, if E is finite over F, then $E = F(\alpha_1, \cdots, \alpha_n)$, where $\alpha_i \in \overline{F}$. Let $\text{irr}(\alpha_i, F(\alpha_1, \cdots, \alpha_{i-1}))$ have α_i as one of n_i distinct zeros that are all of a common multiplicity ν_i, by Theorem 51.2. Then

$$[F(\alpha_1, \cdots, \alpha_i) : F(\alpha_1, \cdots, \alpha_{i-1})] = n_i\nu_i = \{F(\alpha_1, \cdots, \alpha_i) : F(\alpha_1, \cdots, \alpha_{i-1})\}\nu_i.$$

By Theorem 31.4 and Corollary 49.10,

$$[E : F] = \prod_i n_i\nu_i,$$

and

$$\{E : F\} = \prod_i n_i.$$

Therefore, $\{E : F\}$ divides $[E : F]$. ◆

Separable Extensions

51.7 Definition A finite extension E of F is a **separable extension of** F if $\{E : F\} = [E : F]$. An element α of $\overline{F}$ is **separable over** F if $F(\alpha)$ is a separable extension of F. An irreducible polynomial $f(x) \in F[x]$ is **separable over** F if every zero of $f(x)$ in $\overline{F}$ is separable over F. ■

51.8 Example The field $E = \mathbb{Q}[\sqrt{2}, \sqrt{3}]$ is separable over $\mathbb{Q}$ since we saw in Example 50.8 that $\{E : \mathbb{Q}\} = 4 = [E : \mathbb{Q}]$. ▲

To make things a little easier, we have restricted our definition of a separable extension of a field F to *finite* extensions E of F. For the corresponding definition for infinite extensions, see Exercise 12.

We know that $\{F(\alpha) : F\}$ is the number of distinct zeros of $\text{irr}(\alpha, F)$. Also, the multiplicity of α in $\text{irr}(\alpha, F)$ is the same as the multiplicity of each conjugate of α over F, by Theorem 51.2. *Thus α is separable over F if and only if* $\text{irr}(\alpha, F)$ *has all zeros of multiplicity 1. This tells us at once that an irreducible polynomial $f(x) \in F[x]$ is separable over F if and only if $f(x)$ has all zeros of multiplicity 1.*

51.9 Theorem If K is a finite extension of E and E is a finite extension of F, that is, $F \leq E \leq K$, then K is separable over F if and only if K is separable over E and E is separable over F.

Proof Now

$$[K : F] = [K : E][E : F],$$

and

$$\{K : F\} = \{K : E\}\{E : F\}.$$

Then if K is separable over F, so that $[K : F] = \{K : F\}$, we must have $[K : E] = \{K : E\}$ and $[E : F] = \{E : F\}$, since in each case the index divides the degree, by Theorem 51.6. Thus, if K is separable over F, then K is separable over E and E is separable over F.

For the converse, note that $[K : E] = \{K : E\}$ and $[E : F] = \{E : F\}$ imply that

$$[K : F] = [K : E][E : F] = \{K : E\}\{E : F\} = \{K : F\}. \quad \blacklozenge$$

Theorem 51.9 can be extended in the obvious way, by induction, to any finite tower of finite extensions. The top field is a separable extension of the bottom one if and only if each field is a separable extension of the one immediately under it.

51.10 Corollary If E is a finite extension of F, then E is separable over F if and only if each α in E is separable over F.

Proof Suppose that E is separable over F, and let $\alpha \in E$. Then

$$F \leq F(\alpha) \leq E,$$

and Theorem 51.9 shows that $F(\alpha)$ is separable over F.

Suppose, conversely, that every $\alpha \in E$ is separable over F. Since E is a finite extension of F, there exist $\alpha_1, \cdots, \alpha_n$ such that

$$F < F(\alpha_1) < F(\alpha_1, \alpha_2) < \cdots < E = F(\alpha_1, \cdots, \alpha_n).$$

Now since α_i is separable over F, α_i is separable over $F(\alpha_1, \cdots, \alpha_{i-1})$, because

$$q(x) = \text{irr}(\alpha_i, F(\alpha_1, \cdots, \alpha_{i-1}))$$

divides $\text{irr}(\alpha_i, F)$, so that α_i is a zero of $q(x)$ of multiplicity 1. Thus $F(\alpha_1, \cdots, \alpha_i)$ is separable over $F(\alpha_1, \cdots, \alpha_{i-1})$, so E is separable over F by Theorem 51.9, extended by induction. $\blacklozenge$

Perfect Fields

We now turn to the task of proving that α can fail to be separable over F only if F is an infinite field of characteristic $p \neq 0$. One method is to introduce formal derivatives of polynomials. While this is an elegant technique, and also a useful one, we shall, for the sake of brevity, use the following lemma instead. Formal derivatives are developed in Exercises 15 through 22.

51.11 Lemma Let $\bar{F}$ be an algebraic closure of F, and let

$$f(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_1x + a_0$$

be any monic polynomial in $\bar{F}[x]$. If $(f(x))^m \in F[x]$ and $m \cdot 1 \neq 0$ in F, then $f(x) \in F[x]$, that is, all $a_i \in F$.

Proof We must show that $a_i \in F$, and we proceed, by induction on r, to show that $a_{n-r} \in F$. For $r = 1$,

$$(f(x))^m = x^{mn} + (m \cdot 1)a_{n-1}x^{mn-1} + \cdots + {a_0}^m.$$

Since $(f(x))^m \in F[x]$, we have, in particular,

$$(m \cdot 1)a_{n-1} \in F.$$

Thus $a_{n-1} \in F$, since $m \cdot 1 \neq 0$ in F.

As induction hypothesis, suppose that $a_{n-r} \in F$ for $r = 1, 2, \cdots, k$. Then the coefficient of $x^{mn-(k+1)}$ in $(f(x))^m$ is of the form

$$(m \cdot 1)a_{n-(k+1)} + g_{k+1}(a_{n-1}, a_{n-2}, \cdots, a_{n-k}),$$

where $g_{k+1}(a_{n-1}, a_{n-2}, \cdots, a_{n-k})$ is a formal polynomial expression in $a_{n-1}, a_{n-2}, \cdots, a_{n-k}$. By the induction hypothesis that we just stated, $g_{k+1}(a_{n-1}, a_{n-2}, \cdots, a_{n-k}) \in F$, so $a_{n-(k+1)} \in F$, since $m \cdot 1 \neq 0$ in F. ◆

We are now in a position to handle fields F of characteristic zero and to show that for a finite extension E of F, we have $\{E : F\} = [E : F]$. By definition, this amounts to proving that every finite extension of a field of characteristic zero is a separable extension. First, we give a definition.

51.12 Definition A field is **perfect** if every finite extension is a separable extension. ■

51.13 Theorem Every field of characteristic zero is perfect.

Proof Let E be a finite extension of a field F of characteristic zero, and let $\alpha \in E$. Then $f(x) = \text{irr}(\alpha, F)$ factors in $\bar{F}[x]$ into $\prod_i (x - \alpha_i)^\nu$, where the α_i are the distinct zeros of $\text{irr}(\alpha, F)$, and, say, $\alpha = \alpha_1$. Thus

$$f(x) = \left(\prod_i (x - \alpha_i)\right)^\nu,$$

and since $\nu \cdot 1 \neq 0$ for a field F of characteristic 0, we must have

$$\left(\prod_i (x - \alpha_i)\right) \in F[x]$$

by Lemma 51.11. Since $f(x)$ is irreducible and of minimal degree in $F[x]$ having α as a zero, we then see that $\nu = 1$. Therefore, α is separable over F for all $\alpha \in E$. By Corollary 51.10, this means that E is a separable extension of F. ◆

Lemma 54.11 will also get us through for the case of a finite field, although the proof is a bit harder.

51.14 Theorem Every finite field is perfect.

Proof Let F be a finite field of characteristic p, and let E be a finite extension of F. Let $\alpha \in E$. We need to show that α is separable over F. Now $f(x) = \text{irr}(\alpha, F)$ factors in $\bar{F}$ into $\prod_i (x - \alpha_i)^\nu$, where the α_i are the distinct zeros of $f(x)$, and, say, $\alpha = \alpha_1$. Let $\nu = p^t e$,

where p does not divide e. Then

$$f(x) = \prod_i (x - \alpha_i)^{\nu} = \left(\prod_i (x - \alpha_i)^{p^t} \right)^e$$

is in $F[x]$, and by Lemma 54.11, $\prod_i (x - \alpha_i)^{p^t}$ is in $F[x]$ since $e \cdot 1 \neq 0$ in F. Since $f(x) = \text{irr}(\alpha, F)$ is of minimal degree over F having α as a zero, we must have $e = 1$.

Theorem 48.19 and the remark following it show then that

$$f(x) = \prod_i (x - \alpha_i)^{p^t} = \prod_i \left(x^{p^t} - \alpha_i^{p^t} \right).$$

Thus, if we regard $f(x)$ as $g(x^{p^t})$, we must have $g(x) \in F[x]$. Now $g(x)$ is separable over F with distinct zeros $\alpha_i^{p^t}$. Consider $F(\alpha_1^{p^t}) = F(\alpha^{p^t})$. Then $F(\alpha^{p^t})$ is separable over F. Since $x^{p^t} - \alpha^{p^t} = (x - \alpha)^{p^t}$, we see that α is the only zero of $x^{p^t} - \alpha^{p^t}$ in $\overline{F}$. As a finite-dimensional vector space over a finite field F, $F(\alpha^{p^t})$ must be again a finite field. Hence the map

$$\sigma_p : F(\alpha^{p^t}) \to F(\alpha^{p^t})$$

given by $\sigma_p(a) = a^p$ for $a \in F(\alpha^{p^t})$ is an automorphism of $F(\alpha^{p^t})$ by Theorem 48.19. Consequently, $(\sigma_p)^t$ is also an automorphism of $F(\alpha^{p^t})$, and

$$(\sigma_p)^t(a) = a^{p^t}.$$

Since an automorphism of $F(\alpha^{p^t})$ is an onto map, there is $\beta \in F(\alpha^{p^t})$ such that $(\sigma_p)^t(\beta) = \alpha^{p^t}$. But then $\beta^{p^t} = \alpha^{p^t}$, and we saw that α was the only zero of $x^{p^t} - \alpha^{p^t}$, so we must have $\beta = \alpha$. Since $\beta \in F(\alpha^{p^t})$, we have $F(\alpha) = F(\alpha^{p^t})$. Since $F(\alpha^{p^t})$ was separable over F, we now see that $F(\alpha)$ is separable over F. Therefore, α is separable over F and $t = 0$.

We have shown that for $\alpha \in E$, α is separable over F. Then by Corollary 51.10, E is a separable extension of F. ◆

We have completed our aim, which was to show that fields of characteristic 0 and finite fields have only separable finite extensions, that is, these fields are perfect. *For finite extensions E of such perfect fields F, we then have* $[E : F] = \{E : F\}$.

The Primitive Element Theorem

The following theorem is a classic of field theory.

51.15 Theorem **(Primitive Element Theorem)** Let E be a finite separable extension of a field F. Then there exists $\alpha \in E$ such that $F = F(\alpha)$. (Such an element α is a **primitive element.**) That is, a finite separable extension of a field is a simple extension.

Proof If F is a finite field, then E is also finite. Let α be a generator for the cylic group E^* of nonzero elements of E under multiplication. (See Theorem 33.5.) Clearly, $E = F(\alpha)$, so α is a primitive element in this case.

We now assume that F is infinite, and prove our theorem in the case that $E = F(\beta, \gamma)$. The induction argument from this to the general case is straightforward. Let $\text{irr}(\beta, F)$ have distinct zeros $\beta = \beta_1, \cdots, \beta_n$, and let $\text{irr}(\gamma, F)$ have distinct zeros $\gamma = \gamma_1, \cdots, \gamma_m$ in $\overline{F}$, where all zeros are of multiplicity 1, since E is a separable extension of F. Since F is infinite, we can find $a \in F$ such that

$$a \neq (\beta_i - \beta)/(\gamma - \gamma_j)$$

for all i and j, with $j \neq 1$. That is, $a(\gamma - \gamma_j) \neq \beta_i - \beta$. Letting $\alpha = \beta + a\gamma$, we have $\alpha = \beta + a\gamma \neq \beta_i + a\gamma_j$, so

$$\alpha - a\gamma_j \neq \beta_i$$

for all i and all $j \neq 1$. Let $f(x) = \text{irr}(\beta, F)$, and consider

$$h(x) = f(\alpha - ax) \in (F(\alpha))[x].$$

Now $h(\gamma) = f(\beta) = 0$. However, $h(\gamma_j) \neq 0$ for $j \neq 1$ by construction, since the β_i were the only zeros of $f(x)$. Hence $h(x)$ and $g(x) = \text{irr}(\gamma, F)$ have a common factor in $(F(\alpha))[x]$, namely $\text{irr}(\gamma, F(\alpha))$, which must be linear, since γ is the only common zero of $g(x)$ and $h(x)$. Thus $\gamma \in F(\alpha)$, and therefore $\beta = \alpha - a\gamma$ is in $F(\alpha)$. Hence $F(\beta, \gamma) = F(\alpha)$. ◆

51.16 Corollary A finite extension of a field of characteristic zero is a simple extension.

Proof This corollary follows at once from Theorems 51.13 and 51.15. ◆

We see that the only possible "bad case" where a finite extension may not be simple is a finite extension of an infinite field of characteristic $p \neq 0$.

EXERCISES 51

Computations

In Exercises 1 through 4, find α such that the given field is $\mathbb{Q}(\alpha)$. Show that your α is indeed in the given field. Verify by direct computation that the given generators for the extension of $\mathbb{Q}$ can indeed be expressed as formal polynomials in your α with coefficients in $\mathbb{Q}$.

1. $\mathbb{Q}(\sqrt{2}, \sqrt[3]{2})$
2. $\mathbb{Q}(\sqrt[4]{2}, \sqrt[6]{2})$
3. $\mathbb{Q}(\sqrt{2}, \sqrt{3})$
4. $\mathbb{Q}(i, \sqrt[3]{2})$

Concepts

In Exercises 5 and 6, correct the definition of the italicized term without reference to the text, if correction is needed, so that it is in a form acceptable for publication.

5. Let $\overline{F}$ be an algebraic closure of a field F. The *multiplicity of a zero* $\alpha \in \overline{F}$ of a polynomial $f(x) \in F[x]$ is $\nu \in \mathbb{Z}^+$ if and only if $(x - \alpha)^\nu$ is the highest power of $x - \alpha$ that is a factor of $f(x)$ in $F[x]$.
6. Let $\overline{F}$ be an algebraic closure of a field F. An element α in $\overline{F}$ is *separable over* F if and only if α is a zero of multiplicity 1 of $\text{irr}(\alpha, F)$.

7. Give an example of an $f(x) \in \mathbb{Q}[x]$ that has no zeros in $\mathbb{Q}$ but whose zeros in $\mathbb{C}$ are all of multiplicity 2. Explain how this is consistent with Theorem 51.13, which shows that $\mathbb{Q}$ is perfect.
8. Mark each of the following true or false.

   _____ **a.** Every finite extension of every field F is separable over F.
   _____ **b.** Every finite extension of every finite field F is separable over F.
   _____ **c.** Every field of characteristic 0 is perfect.
   _____ **d.** Every polynomial of degree n over every field F always has n distinct zeros in $\bar{F}$.
   _____ **e.** Every polynomial of degree n over every perfect field F always has n distinct zeros in $\bar{F}$.
   _____ **f.** Every irreducible polynomial of degree n over every perfect field F always has n distinct zeros in $\bar{F}$.
   _____ **g.** Every algebraically closed field is perfect.
   _____ **h.** Every field F has an algebraic extension E that is perfect.
   _____ **i.** If E is a finite separable splitting field extension of F, then $|G(E/F)| = [E : F]$.
   _____ **j.** If E is a finite splitting field extension of F, then $|G(E/F)|$ divides $[E : F]$.

Theory

9. Show that if $\alpha, \beta \in \bar{F}$ are both separable over F, then $\alpha \pm \beta$, $\alpha\beta$, and α/β, if $\beta \neq 0$, are all separable over F. [*Hint:* Use Theorem 51.9 and its corollary.]
10. Show that $\{1, y, \cdots, y^{p-1}\}$ is a basis for $\mathbb{Z}_p(y)$ over $\mathbb{Z}_p(y^p)$, where y is an indeterminate. Referring to Example 51.4, conclude by a degree argument that $x^p - t$ is irreducible over $\mathbb{Z}_p(t)$, where $t = y^p$.
11. Prove that if E is an algebraic extension of a perfect field F, then E is perfect.
12. A (possibly infinite) algebraic extension E of a field F is a **separable extension of** F if for every $\alpha \in E$, $F(\alpha)$ is a separable extension of F, in the sense defined in the text. Show that if E is a (possibly infinite) separable extension of F and K is a (possibly infinite) separable extension of E, then K is a separable extension of F.
13. Let E be an algebraic extension of a field F. Show that the set of all elements in E that are separable over F forms a subfield of E, the **separable closure of** F in E. [*Hint:* Use Exercise 9.]
14. Let E be a finite field of order p^n.

    a. Show that the Frobenius automorphism σ_p has order n.
    b. Deduce from part (a) that $G(E/\mathbb{Z}_p)$ is cyclic of order n with generator σ_p. [*Hint:* Remember that

$$|G(E/F)| = \{E : F\} = [E : F]$$

    for a finite separable splitting field extension E over F.]

Exercises 15 through 22 introduce formal derivatives in $F[x]$.

15. Let F be any field and let $f(x) = a_0 + a_1x + \cdots + a_ix^i + \cdots + a_nx^n$ be in $F[x]$. The **derivative** $f'(x)$ **of** $f(x)$ is the polynomial

$$f'(x) = a_1 + \cdots + (i \cdot 1)a_ix^{i-1} + \cdots + (n \cdot 1)a_nx^{n-1},$$

    where $i \cdot 1$ has its usual meaning for $i \in \mathbb{Z}^+$ and $1 \in F$. *These are formal derivatives; no "limits" are involved here.*

    a. Prove that the map $D : F[x] \to F[x]$ given by $D(f(x)) = f'(x)$ is a homomorphism of $\langle F[x], + \rangle$.
    b. Find the kernel of D in the case that F is of characteristic 0.
    c. Find the kernel of D in the case that F is of characteristic $p \neq 0$.

16. Continuing the ideas of Exercise 15, shows that:

a. $D(af(x)) = aD(f(x))$ for all $f(x) \in F[x]$ and $a \in F$.

b. $D(f(x)g(x)) = f(x)g'(x) + f'(x)g(x)$ for all $f(x), g(x) \in F[x]$. [*Hint:* Use part (a) of this exercise and the preceding exercise and proceed by induction on the degree of $f(x)g(x)$.]

c. $D((f(x))^m) = (m \cdot 1)f(x)^{m-1}f'(x)$ for all $f(x) \in F[x]$. [*Hint:* Use part (b).]

17. Let $f(x) \in F[x]$, and let $a \in \bar{F}$ be a zero of $f(x)$ of multiplicity ν. Show that $\nu > 1$ if and only if α is also a zero of $f'(x)$. [*Hint:* Apply parts (b) and (c) of Exercise 16 to the factorization $f(x) = (x - \alpha)^{\nu}g(x)$ of $f(x)$ in the ring $\bar{F}[x]$.]

18. Show from Exercise 17 that every irreducible polynomial over a field F of characteristic 0 is separable. [*Hint:* Use the fact that $\text{irr}(\alpha, F)$ is the *minimal* polynomial for α over F.]

19. Show from Exercise 17 that an irreducible polynomial $q(x)$ over a field F of characteristic $p \neq 0$ is not separable if and only if each exponent of each term of $q(x)$ is divisible by p.

20. Generalize Exercise 17, showing that $f(x) \in F[x]$ has no zero of multiplicity >1 if and only if $f(x)$ and $f'(x)$ have no common factor in $\bar{F}[x]$ of degree >0.

21. Working a bit harder than in Exercise 20, show that $f(x) \in F[x]$ has no zero of multiplicity >1 if and only if $f(x)$ and $f'(x)$ have no common nonconstant factor in $F[x]$. [*Hint:* Use Theorem 46.9 to show that if 1 is a gcd of $f(x)$ and $f'(x)$ in $F[x]$, it is a gcd of these polynomials in $\bar{F}[x]$ also.]

22. Describe a feasible computational procedure for determining whether $f(x) \in F[x]$ has a zero of multiplicity >1, without actually finding the zeros of $f(x)$. [*Hint:* Use Exercise 21.]

SECTION 52 †TOTALLY INSEPARABLE EXTENSIONS

This section shows that a finite extension E of a field F can be split into two stages: a separable extension K of F, followed by a further extension of K to E that is as far from being separable as one can imagine.

We develop our theory of totally inseparable extensions in a fashion parallel to our development of separable extensions.

52.1 Definition A finite extension E of a field F is a **totally inseparable extension of** F if $\{E : F\} = 1 < [E : F]$. An element α of $\bar{F}$ is **totally inseparable over** F if $F(\alpha)$ is totally inseparable over F. ■

We know that $\{F(\alpha) : F\}$ is the number of distinct zeros of $\text{irr}(\alpha, F)$. Thus α is totally inseparable over F if and only if $\text{irr}(\alpha, F)$ has only one zero that is of multiplicity >1.

52.2 Example Referring to Example 51.4, we see that $\mathbb{Z}_p(y)$ is totally inseparable over $\mathbb{Z}_p(y^p)$, where y is an indeterminate. ▲

52.3 Theorem **(Counterpart of Theorem 51.9)** If K is a finite extension of E, E is a finite extension of F, and $F < E < K$, then K is totally inseparable over F if and only if K is totally inseparable over E and E is totally inseparable over F.

† This section is not used in the remainder of the text.

Proof Since $F < E < K$, we have $[K : E] > 1$ and $[E : F] > 1$. Suppose K is totally inseparable over F. Then $\{K : F\} = 1$, and

$$\{K : F\} = \{K : E\}\{E : F\},$$

so we must have

$$\{K : E\} = 1 < [K : E] \quad \text{and} \quad \{E : F\} = 1 < [E : F].$$

Thus K is totally inseparable over E, and E is totally inseparable over F.

Conversely, if K is totally inseparable over E and E is totally inseparable over F, then

$$\{K : F\} = \{K : E\}\{E : F\} = (1)(1) = 1,$$

and $[K : F] > 1$. Thus K is totally inseparable over F. ◆

Theorem 52.3 can be extended by induction, to any finite proper tower of finite extensions. The top field is a totally inseparable extension of the bottom one if and only if each field is a totally inseparable extension of the one immediately under it.

52.4 Corollary **(Counterpart of the Corollary of Theorem 51.10)** If E is a finite extension of F, then E is totally inseparable over F if and only if each α in $E, \alpha \neq F$, is totally inseparable over F.

Proof Suppose that E is totally inseparable over F, and let $\alpha \in E$, with $\alpha \notin F$. Then

$$F < F(\alpha) \leq E.$$

If $F(\alpha) = E$, we are done, by the definition of α totally inseparable over F. If $F < F(\alpha) < E$, then Theorem 52.3 shows that since E is totally inseparable over F, $F(\alpha)$ is totally inseparable over F.

Conversely, suppose that for every $\alpha \in E$, with $\alpha \notin F, \alpha$ is totally inseparable over F. Since E is finite over F, there exist $\alpha_1, \cdots, \alpha_n$ such that

$$F < F(\alpha_1) < F(\alpha_1, \alpha_2) < \cdots < E = F(\alpha_1, \cdots, \alpha_n).$$

Now since α_i is totally inseparable over F, α_i is totally inseparable over $F(\alpha_1, \cdots, \alpha_{i-1})$, because $q(x) = \text{irr}(\alpha_i, F(\alpha_1, \cdots, \alpha_{i-1}))$ divides $\text{irr}(\alpha_i, F)$ so that α_i is the only zero of $q(x)$ and is of multiplicity >1. Thus $F(\alpha_1, \cdots, \alpha_i)$ is totally inseparable over $F(\alpha_1, \cdots, \alpha_{i-1})$, and E is totally inseparable over F, by Theorem 52.3, extended by induction. ◆

Thus far we have so closely paralleled our work in Section 51 that we could have handled these ideas together.

Separable Closures

We now come to our main reason for including this material.

52.5 Theorem Let F have characteristic $p \neq 0$, and let E be a finite extension of F. Then $\alpha \in E, \alpha \notin F$, is totally inseparable over F if and only if there is some integer $t \geq 1$ such that $\alpha^{p^t} \in F$.

Furthermore, there is a unique extension K of F, with $F \leq K \leq E$, such that K is separable over F, and either $E = K$ or E is totally inseparable over K.

Proof Let $\alpha \in E, \alpha \notin F$, be totally inseparable over F. Then $\text{irr}(\alpha, F)$ has just one zero α of multiplicity >1, and, as shown in the proof of Theorem 51.14, $\text{irr}(\alpha, F)$ must be of the form

$$x^{p^t} - \alpha^{p^t}.$$

Hence $\alpha^{p^t} \in F$ for some $t \geq 1$.

Conversely, if $\alpha^{p^t} \in F$ for some $t \geq 1$, where $\alpha \in E$ and $\alpha \notin F$, then

$$x^{p^t} - \alpha^{p^t} = (x - \alpha)^{p^t},$$

and $(x^{p^t} - \alpha^{p^t}) \in F[x]$, showing that $\text{irr}(\alpha, F)$ divides $(x - \alpha)^{p^t}$. Thus $\text{irr}(\alpha, F)$ has α as its only zero and this zero is of multiplicity >1, so α is totally inseparable over F.

For the second part of the theorem, let $E = F(\alpha_1, \cdots, \alpha_n)$. Then if

$$\text{irr}(\alpha_i, F) = \prod_j \left(x^{p^{t_i}} - {\alpha_{ij}}^{p^{t_i}}\right),$$

with $\alpha_{i1} = \alpha_i$, let $\beta_{ij} = {\alpha_{ij}}^{p^{t_i}}$. We have $F(\beta_{11}, \beta_{21}, \cdots, \beta_{n1}) \leq E$, and β_{i1} is a zero of

$$f_i(x) = \prod_j (x - \beta_{ij}),$$

where $f_i(x) \in F[x]$. Now since raising to the power p is an isomorphism σ_p of E onto a subfield of E, raising to the power of p^t is the isomorphic mapping $(\sigma_p)^t$ of E onto a subfield of E. Thus since the α_{ij} are all distinct for a fixed i, so are the β_{ij} for a fixed i. Therefore, β_{ij} is separable over F, because it is a zero of a polynomial $f_i(x)$ in $F[x]$ with zeros of multiplicity 1. Then

$$K = F(\beta_{11}, \beta_{21}, \cdots, \beta_{n1})$$

is separable over F, by the proof of Corollary 51.10. If all $p^{t_i} = 1$, then $K = E$. If some $p^{t_i} \neq 1$, then $K \neq E$, and $\alpha_i^{p^{t_i}} = \beta_{i1}$ is in K, showing that each $\alpha_i \notin K$ is totally inseparable over K, by the first part of this theorem. Hence $E = K(\alpha_1, \cdots, \alpha_n)$ is totally inseparable over K, by the proof of Corollary 52.4.

It follows from Corollaries 51.10 and 52.4 that the field K consists of all elements α in E that are separable over F. Thus K is unique. ◆

52.6 Definition The unique field K of Theorem 52.5 is the **separable closure of F in E.** ■

The preceding theorem shows the precise structure of totally inseparable extensions of a field of characteristic p. Such an extension can be obtained by repeatedly adjoining pth roots of elements that are not already pth powers.

We remark that Theorem 52.5 is true for infinite algebraic extensions E of F. The proof of the first assertion of the theorem is valid for the case of infinite extensions also. For the second part, since $\alpha \pm \beta$, $\alpha\beta$, and α/β, for $\beta \neq 0$, are all contained in the field $F(\alpha, \beta)$, all elements of E separable over F form a subfield K of E, the **separable closure of F in E**. It follows that an $\alpha \in E, \alpha \notin K$, is totally inseparable over K, since α and all coefficients of $\text{irr}(\alpha, K)$ are in a finite extension of F, and then Theorem 52.5 can be applied.

EXERCISES 52

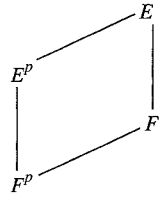
Concepts

1. Let y and z be indeterminates, and let $u = y^{12}$ and $v = z^{18}$. Describe the separable closure of $\mathbb{Z}_3(u, v)$ in $\mathbb{Z}_3(y, z)$.
2. Let y and z be indeterminates, and let $u = y^{12}$ and $v = y^2z^{18}$. Describe the separable closure of $\mathbb{Z}_3(u, v)$ in $\mathbb{Z}_3(y, z)$.
3. Referring to Exercise 1, describe the totally inseparable closure (see Exercise 6) of $\mathbb{Z}_3(u, v)$ in $\mathbb{Z}_3(y, z)$.
4. Referring to Exercise 2, describe the totally inseparable closure of $\mathbb{Z}_3(u, v)$ in $\mathbb{Z}_3(y, z)$. (See Exercise 6.)
5. Mark each of the following true or false.

   _____ **a.** No proper algebraic extension of an infinite field of characteristic $p \neq 0$ is ever a separable extension.
   _____ **b.** If $F(\alpha)$ is totally inseparable over F of characteristic $p \neq 0$, then $\alpha^{p^t} \in F$ for some $t > 0$.
   _____ **c.** For an indeterminate y, $\mathbb{Z}_5(y)$ is separable over $\mathbb{Z}_5(y^5)$.
   _____ **d.** For an indeterminate y, $\mathbb{Z}_5(y)$ is separable over $\mathbb{Z}_5(y^{10})$.
   _____ **e.** For an indeterminate y, $\mathbb{Z}_5(y)$ is totally inseparable over $\mathbb{Z}_5(y^{10})$.
   _____ **f.** If F is a field and α is algebraic over F, then α is either separable or totally inseparable over F.
   _____ **g.** If E is an algebraic extension of a field F, then F has a separable closure in E.
   _____ **h.** If E is an algebraic extension of a field F, then E is totally inseparable over the separable closure of F in E.
   _____ **i.** If E is an algebraic extension of a field F and E is not a separable extension of F, then E is totally inseparable over the separable closure of F in E.
   _____ **j.** If α is totally inseparable over F, then α is the only zero of $\text{irr}(\alpha, F)$.

Theory

6. Show that if E is an algebraic extension of a field F, then the union of F with the set of all elements of E totally inseparable over F forms a subfield of E, the **totally inseparable closure of F in E**.
7. Show that a field F of characteristic $p \neq 0$ is perfect if and only if $F^p = F$, that is, every element of F is a pth power of some element of F.
8. Let E be a finite extension of a field F of characteristic p. In the notation of Exercise 7, show that $E^p = E$ if and only if $F^p = F$. [*Hint:* The map $\sigma_p : E \to E$ defined by $\sigma_p(\alpha) = \alpha^p$ for $\alpha \in E$ is an isomorphism onto a subfield of E. Consider the diagram in Fig. 52.7, and make degree arguments.]



52.7 Figure

SECTION 53 GALOIS THEORY

Résumé

This section is perhaps the climax in elegance of the subject matter of the entire text. The Galois theory gives a beautiful interplay of group and field theory. Starting with Section 48, our work has been aimed at this goal. We shall start by recalling the main results we have developed and should have well in mind.

1. Let $F \leq E \leq \bar{F}$, $\alpha \in E$, and let β be a conjugate of α over F, that is, $\text{irr}(\alpha, F)$ has β as a zero also. Then there is an isomorphism $\psi_{\alpha,\beta}$ mapping $F(\alpha)$ onto $F(\beta)$ that leaves F fixed and maps α onto β.
2. If $F \leq E \leq \bar{F}$ and $\alpha \in E$, then an automorphism σ of $\bar{F}$ that leaves F fixed *must* map α onto some conjugate of α over F.
3. If $F \leq E$, the collection of all automorphisms of E leaving F fixed forms a group $G(E/F)$. For any subset S of $G(E/F)$, the set of all elements of E left fixed by all elements of S is a field E_S. Also, $F \leq E_{G(E/F)}$.
4. A field E, $F \leq E \leq \bar{F}$, is a splitting field over F if and only if every isomorphism of E onto a subfield of $\bar{F}$ leaving F fixed is an automorphism of E. If E is a finite extension and a splitting field over F, then $|G(E/F)| = \{E : F\}$.
5. If E is a finite extension of F, then $\{E : F\}$ divides $[E : F]$. If E is also separable over F, then $\{E : F\} = [E : F]$. Also, E is separable over F if and only if $\text{irr}(\alpha, F)$ has all zeros of multiplicity 1 for every $\alpha \in E$.
6. If E is a finite extension of F and is a separable splitting field over F, then $|G(E/F)| = \{E : F\} = [E : F]$.

Normal Extensions

We are going to be interested in finite extensions K of F such that every isomorphism of K onto a subfield of $\bar{F}$ leaving F fixed is an automorphism of K and such that

$$[K : F] = \{K : F\}.$$

In view of results 4 and 5, these are the finite extensions of F that are separable splitting fields over F.

53.1 Definition A finite extension K of F is a **finite normal extension of** F if K is a separable splitting field over F. ■

Suppose that K is a finite normal extension of F, where $K \leq \bar{F}$, as usual. Then by result 4, every automorphism of $\bar{F}$ leaving F fixed induces an automorphism of K. As before, we let $G(K/F)$ be the group of all automorphisms of K leaving F fixed. After one more result, we shall be ready to illustrate the main theorem.

53.2 Theorem Let K be a finite normal extension of F, and let E be an extension of F, where $F \leq E \leq K \leq \bar{F}$. Then K is a finite normal extension of E, and $G(K/E)$ is precisely the subgroup

of $G(K/F)$ consisting of all those automorphisms that leave E fixed. Moreover, two automorphisms σ and τ in $G(K/F)$ induce the same isomorphism of E onto a subfield of $\bar{F}$ if and only if they are in the same left coset of $G(K/E)$ in $G(K/F)$.

Proof If K is the splitting field of a set $\{f_i(x) \mid i \in I\}$ of polynomials in $F[x]$, then K is the splitting field over E of this same set of polynomials viewed as elements of $E[x]$. Theorem 51.9 shows that K is separable over E, since K is separable over F. Thus K is a normal extension of E. This establishes our first contention.

Now every element of $G(K/E)$ is an automorphism of K leaving F fixed, since it even leaves the possibly larger field E fixed. Thus $G(K/E)$ can be viewed as a subset of $G(K/F)$. Since $G(K/E)$ is a group under function composition also, we see that $G(K/E) \leq G(K/F)$.

Finally, for σ and τ in $G(K/F)$, σ and τ are in the same left coset of $G(K/E)$ if and only if $\tau^{-1}\sigma \in G(K/E)$ or if and only if $\sigma = \tau\mu$ for $\mu \in G(K/E)$. But if $\sigma = \tau\mu$ for $\mu \in G(K/E)$, then for $\alpha \in E$, we have

$$\sigma(\alpha) = (\tau\mu)(\alpha) = \tau(\mu(\alpha)) = \tau(\alpha),$$

since $\mu(\alpha) = \alpha$ for $\alpha \in E$. Conversely, if $\sigma(\alpha) = \tau(\alpha)$ for all $\alpha \in E$, then

$$(\tau^{-1}\sigma)(\alpha) = \alpha$$

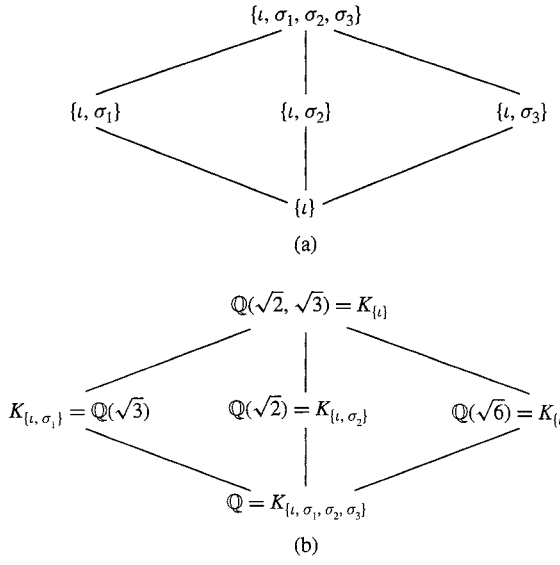
for all $\alpha \in E$, so $\tau^{-1}\sigma$ leaves E fixed, and $\mu = \tau^{-1}\sigma$ is thus in $G(K/E)$. ◆

The preceding theorem shows that there is a one-to-one correspondence between left cosets of $G(K/E)$ in $G(K/F)$ and isomorphisms of E onto a subfield of K leaving F fixed. Note that we cannot say that these left cosets correspond to *automorphisms* of E over F, since E may not be a splitting field over F. Of course, if E is a *normal* extension of F, then these isomorphisms would be automorphisms of E over F. We might guess that this will happen if and only if $G(K/E)$ is a *normal* subgroup of $G(K/F)$, and this is indeed the case. That is, the two different uses of the word *normal* are really closely related. Thus if E is a normal extension of F, then the left cosets of $G(K/E)$ in $G(K/F)$ can be viewed as elements of the *factor group* $G(K/F)/G(K/E)$, which is then a group of automorphisms acting on E and leaving F fixed. We shall show that this factor group is isomorphic to $G(E/F)$.

The Main Theorem

The Main Theorem of Galois Theory states that for a finite normal extension K of a field F, there is a one-to-one correspondence between the subgroups of $G(K/F)$ and the intermediate fields E, where $F \leq E \leq K$. *This correspondence associates with each intermediate field E the subgroup $G(K/E)$. We can also go the other way and start with a subgroup H of $G(K/F)$ and associate with H its fixed field K_H.* We shall illustrate this with an example, then state the theorem and discuss its proof.

53.3 Example Let $K = \mathbb{Q}(\sqrt{2}, \sqrt{3})$. Now K is a normal extension of $\mathbb{Q}$, and Example 48.17 showed that there are four automorphisms of K leaving $\mathbb{Q}$ fixed. We recall them by giving their values on the basis $\{1, \sqrt{2}, \sqrt{3}, \sqrt{6}\}$ for K over $\mathbb{Q}$.



53.4 Figure (a) Group diagram. (b) Field diagram.

ι : The identity map
σ_1 : Maps $\sqrt{2}$ onto $-\sqrt{2}$, $\sqrt{6}$ onto $-\sqrt{6}$, and leaves the others fixed
σ_2 : Maps $\sqrt{3}$ onto $-\sqrt{3}$, $\sqrt{6}$ onto $-\sqrt{6}$, and leaves the others fixed
σ_3 : Maps $\sqrt{2}$ onto $-\sqrt{2}$, $\sqrt{3}$ onto $-\sqrt{3}$, and leaves the others fixed

We saw that $\{\iota, \sigma_1, \sigma_2, \sigma_3\}$ is isomorphic to the Klein 4-group. The complete list of subgroups, with each subgroup paired off with the corresponding intermediate field that it leaves fixed, is as follows:

$$\begin{aligned} \{\iota, \sigma_1, \sigma_2, \sigma_3\} &\leftrightarrow \mathbb{Q}, \\ \{\iota, \sigma_1\} &\leftrightarrow \mathbb{Q}(\sqrt{3}), \\ \{\iota, \sigma_2\} &\leftrightarrow \mathbb{Q}(\sqrt{2}), \\ \{\iota, \sigma_3\} &\leftrightarrow \mathbb{Q}(\sqrt{6}), \\ \{\iota\} &\leftrightarrow \mathbb{Q}(\sqrt{2}, \sqrt{3}). \end{aligned}$$

All subgroups of the abelian group $\{\iota, \sigma_1, \sigma_2, \sigma_3\}$ are normal subgroups, and all the intermediate fields are normal extensions of $\mathbb{Q}$. Isn't that elegant?

Note that if one subgroup is contained in another, then the larger of the two subgroups corresponds to the smaller of the two corresponding fixed fields. The larger the subgroup, that is, the more automorphisms, the smaller the fixed field, that is, the fewer elements left fixed. In Fig. 53.4 we give the corresponding diagrams for the subgroups and intermediate fields. *Note again that the groups near the top correspond to the fields near the bottom.* That is, one diagram looks like the other *inverted* or turned upside down. Since here each diagram actually looks like itself turned upside down, this is not a good example for us

to use to illustrate this *inversion principle*. Turn ahead to Fig. 54.6 to see diagrams that do not look like their own inversions. ▲

53.5 Definition If K is a finite normal extension of a field F, then $G(K/F)$ is the **Galois group of K over F.** ■

We shall now state the main theorem, then give another example, and finally, complete the proof of the main theorem.

53.6 Theorem **(Main Theorem of Galois Theory)** Let K be a finite normal extension of a field F, with Galois group $G(K/F)$. For a field E, where $F \leq E \leq K$, let $\lambda(E)$ be the subgroup of $G(K/F)$ leaving E fixed. Then λ is a one-to-one map of the set of all such intermediate fields E onto the set of all subgroups of $G(K/F)$. The following properties hold for λ:

1. $\lambda(E) = G(K/E)$.
2. $E = K_{G(K/E)} = K_{\lambda(E)}$.
3. For $H \leq G(K/F)$, $\lambda(E_H) = H$.
4. $[K : E] = |\lambda(E)|$ and $[E : F] = (G(K/F) : \lambda(E))$, the number of left cosets of $\lambda(E)$ in $G(K/F)$.
5. E is a normal extension of F if and only if $\lambda(E)$ is a normal subgroup of $G(K/F)$. When $\lambda(E)$ is a normal subgroup of $G(K/F)$, then
$$G(E/F) \simeq G(K/F)/G(K/E).$$
6. The diagram of subgroups of $G(K/F)$ is the inverted diagram of intermediate fields of K over F.

Observations on the Proof We have really already proved a substantial part of this theorem. Let us see just how much we have left to prove.

Property 1 is just the definition of λ found in the statement of the theorem. For Property 2, Theorem 48.15 shows that
$$E \leq K_{G(K/E)}.$$
Let $\alpha \in K$, where $\alpha \notin E$. Since K is a normal extension of E, by using a conjugation isomorphism and the Isomorphism Extension Theorem, we can find an automorphism of K leaving E fixed and mapping α onto a different zero of $\text{irr}(\alpha, F)$. This implies that
$$K_{G(K/E)} \leq E,$$
so $E = K_{G(K/E)}$. This disposes of Property 2 and also tells us that λ is one to one, for if $\lambda(E_1) = \lambda(E_2)$, then by Property 2, we have
$$E_1 = K_{\lambda(E_1)} = K_{\lambda(E_2)} = E_2.$$

Now Property 3 is going to be our main job. This amounts exactly to showing that λ is an onto map. Of course, for $H \leq G(K/F)$, we have $H \leq \lambda(K_H)$, for H surely is included in the set of all automorphisms leaving K_H fixed. Here we will be using strongly our property $[K : E] = \{K : E\}$.

Property 4 follows from $[K : E] = \{K : E\}$, $[E : F] = \{E : F\}$, and the last statement in Theorem 53.2.

We shall have to show that the two senses of the word *normal* correspond for Property 5.

We have already disposed of Property 6 in Example 53.3. *Thus only Properties 3 and 5 remain to be proved.*

The Main Theorem of Galois Theory is a strong tool in the study of zeros of polynomials. If $f(x) \in F[x]$ is such that every irreducible factor of $f(x)$ is separable over F, then the splitting field K of $f(x)$ over F is a normal extension of F. The Galois group $G(K/F)$ is the **group of the polynomial** $f(x)$ **over** F. The structure of this group may give considerable information regarding the zeros of $f(x)$. This will be strikingly illustrated in Section 56 when we achieve our *final goal.*

Galois Groups over Finite Fields

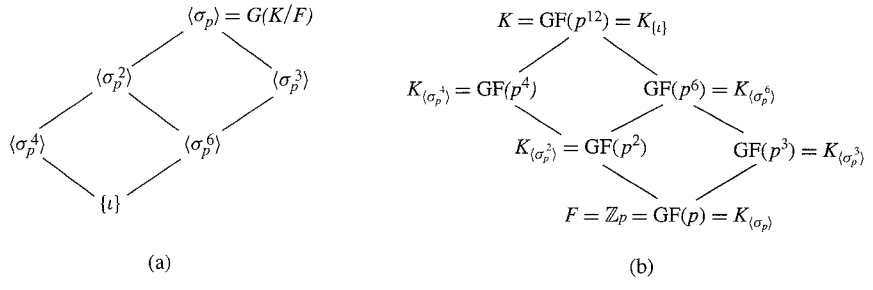
Let K be a finite extension of a *finite field* F. We have seen that K is a separable extension of F (a finite field is perfect). Suppose that the order of F is p^r and $[K : F] = n$, so the order of K is p^{rn}. Then we have seen that K is the splitting field of $x^{p^{rn}} - x$ over F. Hence K is a normal extension of F.

Now one automorphism of K that leaves F fixed is σ_{p^r}, where for $\alpha \in K$, $\sigma_{p^r}(\alpha) = \alpha^{p^r}$. Note that $(\sigma_{p^r})^i(\alpha) = \alpha^{p^{ri}}$. Since a polynomial of degree p^{ri} can have at most p^{ri} zeros in a field, we see that the smallest power of σ_{p^r} that could possibly leave all p^{rn} elements of K fixed is the nth power. That is, the order of the element σ_{p^r} in $G(K/F)$ is at least n. Therefore, since $|G(K/F)| = [K : F] = n$, it must be that $G(K/F)$ is cyclic and generated by σ_{p^r}. We summarize these arguments in a theorem.

53.7 Theorem Let K be a finite extension of degree n of a finite field F of p^r elements. Then $G(K/F)$ is cyclic of order n, and is generated by σ_{p^r}, where for $\alpha \in K$, $\sigma_{p^r}(\alpha) = \alpha^{p^r}$.

We use this theorem to give another illustration of the Main Theorem of Galois Theory.

53.8 Example Let $F = \mathbb{Z}_p$, and let $K = \mathrm{GF}(p^{12})$, so $[K : F] = 12$. Then $G(K/F)$ is isomorphic to the cyclic group $\langle \mathbb{Z}_{12}, + \rangle$. The diagrams for the subgroups and for the intermediate fields are given in Fig. 53.9. Again, each diagram is not only the inversion of the other, but unfortunately, also looks like the inversion of itself. Examples where the diagrams do not look like their own inversion are given in next Section 54. We describe the cyclic



53.9 Figure (a) Group diagram. (b) Field diagram.

subgroups of $G(K/F) = \langle \sigma_p \rangle$ by giving generators, for example,

$$\langle {\sigma_p}^4 \rangle = \{\iota, {\sigma_p}^4, {\sigma_p}^8\}. \quad \blacktriangle$$

Proof of the Main Theorem Completed

We saw that Properties 3 and 5 are all that remain to be proved in the Main Theorem of Galois Theory.

Proof Turning to Property 3, we must show that for $H \leq G(K/F)$, $\lambda(K_H) = H$. We know that $H \leq \lambda(K_H) \leq G(K/F)$. Thus what we really must show is that it is impossible to have H a *proper* subgroup of $\lambda(K_H)$. We shall suppose that

$$H < \lambda(K_H)$$

and shall derive a contradiction. As a finite separable extension, $K = K_H(\alpha)$ for some $\alpha \in K$, by Theorem 51.15. Let

$$n = [K : K_H] = \{K : K_H\} = |G(K/K_H)|.$$

Then $H < G(K/K_H)$ implies that $|H| < |G(K/K_H)| = n$. Thus we would have to have $|H| < [K : K_H] = n$. Let the elements of H be $\sigma_1, \cdots, \sigma_{|H|}$, and consider the polynomial

$$f(x) = \prod_{i=1}^{|H|}(x - \sigma_i(\alpha)).$$

Then $f(x)$ is of degree $|H| < n$. Now the coefficients of each power of x in $f(x)$ are *symmetric* expressions in the $\sigma_i(\alpha)$. For example, the coefficient of $x^{|H|-1}$ is $-\sigma_1(\alpha) - \sigma_2(\alpha) - \cdots - \sigma_{|H|}(\alpha)$. Thus these coefficients are invariant under each isomorphism $\sigma_i \in H$, since if $\sigma \in H$, then

$$\sigma\sigma_1, \cdots, \sigma\sigma_{|H|}$$

is again the sequence $\sigma_1, \cdots, \sigma_{|H|}$, except for order, H being a group. Hence $f(x)$ has coefficients in K_H, and since some σ_i is ι, we see that some $\sigma_i(\alpha)$ is α, so $f(\alpha) = 0$. Therefore, we would have

$$\deg(\alpha, K_H) \leq |H| < n = [K : K_H] = [K_H(\alpha) : K_H].$$

This is impossible. Thus we have proved Property 3.

We turn to Property 5. Every extension E of F, $F \leq E \leq K$, is separable over F, by Theorem 51.9. Thus E is normal over F if and only if E is a splitting field over F. By the Isomorphism Extension Theorem, every isomorphism of E onto a subfield of $\bar{F}$ leaving F fixed can be extended to an *automorphism* of K, since K is *normal* over F. Thus the automorphisms of $G(K/F)$ induce all possible isomorphisms of E onto a subfield of $\bar{F}$ leaving F fixed. By Theorem 50.3, this shows that E is a splitting field over F, and hence is normal over F, if and only if for all $\sigma \in G(K/F)$ and $\alpha \in E$,

$$\sigma(\alpha) \in E.$$

By Property 2, E is the fixed field of $G(K/E)$, so $\sigma(\alpha) \in E$ if and only if for all $\tau \in G(K/E)$

$$\tau(\sigma(\alpha)) = \sigma(\alpha).$$

This in turn holds if and only if

$$(\sigma^{-1}\tau\sigma)(\alpha) = \alpha$$

for all $\alpha \in E$, $\sigma \in G(K/F)$, and $\tau \in G(K/E)$. But this means that for all $\sigma \in G(K/F)$ and $\tau \in G(K/E)$, $\sigma^{-1}\tau\sigma$ leaves every element of E fixed, that is,

$$(\sigma^{-1}\tau\sigma) \in G(K/E).$$

This is precisely the condition that $G(K/F)$ be a normal subgroup of $G(K/F)$.

It remains for us to show that when E is a normal extension of F, $G(E/F) \simeq G(K/F)/G(K/E)$. For $\sigma \in G(K/F)$, let σ_E be the *automorphism* of E induced by σ (we are assuming that E is a *normal* extension of F). Thus $\sigma_E \in G(E/F)$. The map $\phi : G(K/F) \to G(K/F)$ given by

$$\phi(\sigma) = \sigma_E$$

for $\sigma \in G(K/F)$ is a homomorphism. By the Isomorphism Extension Theorem, every automorphism of E leaving F fixed can be extended to some automorphism of K; that is, it is τ_E for some $\tau \in G(K/F)$. Thus ϕ is onto $G(E/F)$. The kernel of ϕ is $G(K/E)$. Therefore, by the Fundamental Isomorphism Theorem, $G(E/F) \simeq G(K/F)/G(K/E)$. Furthermore, this isomorphism is a natural one. ◆

EXERCISES 53

Computations

The field $K = \mathbb{Q}(\sqrt{2}, \sqrt{3}, \sqrt{5})$ is a finite normal extension of $\mathbb{Q}$. It can be shown that $[K : \mathbb{Q}] = 8$. In Exercises 1 through 8, compute the indicated numerical quantity. The notation is that of Theorem 53.6.

1. $\{K : \mathbb{Q}\}$

2. $|G(K/\mathbb{Q})|$

3. $|\lambda(\mathbb{Q})|$

4. $|\lambda(\mathbb{Q}(\sqrt{2}, \sqrt{3}))|$

5. $|\lambda(\mathbb{Q}(\sqrt{6}))|$

6. $|\lambda(\mathbb{Q}(\sqrt{30}))|$

7. $|\lambda(\mathbb{Q}(\sqrt{2} + \sqrt{6}))|$

8. $|\lambda(K)|$

9. Describe the group of the polynomial $(x^4 - 1) \in \mathbb{Q}[x]$ over $\mathbb{Q}$.

10. Give the order and describe a generator of the group $G(\mathrm{GF}(729)/\mathrm{GF}(9))$.

11. Let K be the splitting field of $x^3 - 2$ over $\mathbb{Q}$. (Refer to Example 50.9.)

a. Describe the six elements of $G(K/\mathbb{Q})$ by giving their values on $\sqrt[3]{2}$ and $i\sqrt{3}$. (By Example 50.9, $K = \mathbb{Q}(\sqrt[3]{2}, i\sqrt{3})$.)

b. To what group we have seen before is $G(K/\mathbb{Q})$ isomorphic?

c. Using the notation given in the answer to part (a) in the back of the text, give the diagrams for the subfields of K and for the subgroups of $G(K/\mathbb{Q})$, indicating corresponding intermediate fields and subgroups, as we did in Fig. 53.4.

12. Describe the group of the polynomial $(x^4 - 5x^2 + 6) \in \mathbb{Q}[x]$ over $\mathbb{Q}$.

13. Describe the group of the polynomial $(x^3 - 1) \in \mathbb{Q}[x]$ over $\mathbb{Q}$.

Concepts

14. Give an example of two finite normal extensions K_1 and K_2 of the same field F such that K_1 and K_2 are not isomorphic fields but $G(K_1/F) \simeq G(K_2/F)$.

15. Mark each of the following true or false.

_____ **a.** Two different subgroups of a Galois group may have the same fixed field.

_____ **b.** In the notation of Theorem 53.6, if $F \leq E < L \leq K$, then $\lambda(E) < \lambda(L)$.

_____ **c.** If K is a finite normal extension of F, then K is a normal extension of E, where $F \leq E \leq K$.

_____ **d.** If two finite normal extensions E and L of a field F have isomorphic Galois groups, then $[E : F] = [L : F]$.

_____ **e.** If E is a finite normal extension of F and H is a normal subgroup of $G(E/F)$, then E_H is a normal extension of F.

_____ **f.** If E is any finite normal simple extension of a field F, then the Galois group $G(E/F)$ is a simple group.

_____ **g.** No Galois group is simple.

_____ **h.** The Galois group of a finite extension of a finite field is abelian.

_____ **i.** An extension E of degree 2 over a field F is always a normal extension of F.

_____ **j.** An extension E of degree 2 over a field F is always a normal extension of F if the characteristic of F is not 2.

Theory

16. A finite normal extension K of a field F is **abelian over** F if $G(K/F)$ is an abelian group. Show that if K is abelian over F and E is a normal extension of F, where $F \leq E \leq K$, then K is abelian over E and E is abelian over F.

17. Let K be a finite normal extension of a field F. Prove that for every $\alpha \in K$, the **norm of** α **over** F, given by

$$N_{K/F}(\alpha) = \prod_{\sigma \in G(K/F)} \sigma(\alpha),$$

and the **trace of** α **over** F, given by

$$Tr_{K/F}(\alpha) = \sum_{\sigma \in G(K/F)} \sigma(\alpha),$$

are elements of F.

18. Consider $K = \mathbb{Q}(\sqrt{2}, \sqrt{3})$. Referring to Exercise 17, compute each of the following (see Example 53.3).

a. $N_{K/\mathbb{Q}}(\sqrt{2})$ **b.** $N_{K/\mathbb{Q}}(\sqrt{2}+\sqrt{3})$

c. $N_{K/\mathbb{Q}}(\sqrt{6})$ **d.** $N_{K/\mathbb{Q}}(2)$

e. $Tr_{K/\mathbb{Q}}(\sqrt{2})$ **f.** $Tr_{K/\mathbb{Q}}(\sqrt{2}+\sqrt{3})$

g. $Tr_{K/\mathbb{Q}}(\sqrt{6})$ **h.** $Tr_{K/\mathbb{Q}}(2)$

19. Let K be a normal extension of F, and let $K = F(\alpha)$. Let

$$\text{irr}(\alpha, F) = x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0.$$

Referring to Exercise 17, show that

a. $N_{K/F}(\alpha) = (-1)^n a_0$, **b.** $Tr_{K/F}(\alpha) = -a_{n-1}$.

20. Let $f(x) \in F[x]$ be a polynomial of degree n such that each irreducible factor is separable over F. Show that the order of the group of $f(x)$ over F divides $n!$.

21. Let $f(x) \in F[x]$ be a polynomial such that every irreducible factor of $f(x)$ is a separable polynomial over F. Show that the group of $f(x)$ over F can be viewed in a natural way as a group of permutations of the zeros of $f(x)$ in $\bar{F}$.

22. Let F be a field and let ζ be a primitive nth root of unity in $\bar{F}$, where the characteristic of F is either 0 or does not divide n.

a. Show that $F(\zeta)$ is a normal extension of F.

b. Show that $G(F(\zeta)/F)$ is abelian. [*Hint:* Every $\sigma \in G(F(\zeta)/F)$ maps ζ onto some ζ^r and is completely determined by this value r.]

23. A finite normal extension K of a field F is **cyclic over** F if $G(K/F)$ is a cyclic group.

a. Show that if K is cyclic over F and E is a normal extension of F, where $F \leq E \leq K$, then E is cyclic over F and K is cyclic over E.

b. Show that if K is cyclic over F, then there exists exactly one field E, $F \leq E \leq K$, of degree d over F for each divisor d of $[K : F]$.

24. Let K be a finite normal extension of F.

a. For $\alpha \in K$, show that

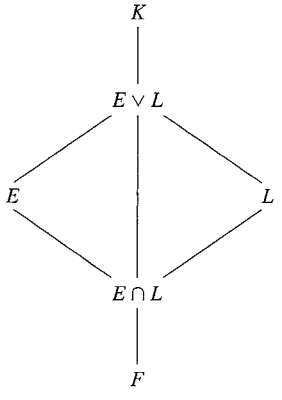
$$f(x) = \prod_{\sigma \in G(K/F)} (x - \sigma(\alpha))$$

is in $F[x]$.

b. Referring to part (a), show that $f(x)$ is a power of $\text{irr}(\alpha, F)$, and $f(x) = \text{irr}(\alpha, F)$ if and only if $K = F(\alpha)$.

25. The **join** $E \vee L$ of two extension fields E and L of F in $\bar{F}$ is the smallest subfield of $\bar{F}$ containing both E and L. That is, $E \vee L$ is the intersection of all subfields of $\bar{F}$ *containing both* E and L. Let K be a finite normal extension of a field F, and let E and L be extensions of F contained in K, as shown in Fig. 53.10. Describe $G(K/(E \vee L))$ in terms of $G(K/E)$ and $G(K/L)$.

26. With reference to the situation in Exercise 25, describe $G\{K/(E \cap L)\}$ in terms of $G(K/E)$ and $G(K/L)$.



53.9 Figure

SECTION 54 ILLUSTRATIONS OF GALOIS THEORY

Symmetric Functions

Let F be a field, and let $y_1, \cdots, y_n$ be indeterminates. There are some natural automorphisms of $F(y_1, \cdots, y_n)$ leaving F fixed, namely, those defined by permutations of $\{y_1, \cdots, y_n\}$. To be more explicit, let σ be a permutation of $\{1, \cdots, n\}$, that is, $\sigma \in S_n$. Then σ gives rise to a natural map $\overline{\sigma} : F(y_1, \cdots, y_n) \to F(y_1, \cdots, y_n)$ given by

$$\overline{\sigma}\left(\frac{f(y_1, \cdots, y_n)}{g(y_1, \cdots, y_n)}\right) = \frac{f(y_{\sigma(1)}, \cdots, y_{\sigma(n)})}{g(y_{\sigma(1)}, \cdots, y_{\sigma(n)})}$$

for $f(y_1, \cdots, y_n), g(y_1, \cdots, y_n) \in F[y_1, \cdots, y_n]$, with $g(y_1, \cdots, y_n) \neq 0$. It is immediate that $\overline{\sigma}$ is an automorphism of $F(y_1, \cdots, y_n)$ leaving F fixed. The elements of $F(y_1, \cdots, y_n)$ left fixed by *all* $\overline{\sigma}$, for all $\sigma \in S_n$, are those rational functions that are *symmetric* in the indeterminates $y_1, \cdots, y_n$.

54.1 Definition An element of the field $F(y_1, \cdots, y_n)$ is a **symmetric function in** $y_1, \cdots, y_n$ **over** F, if it is left fixed by all permutations of $y_1, \cdots, y_n$, in the sense just explained. ■

Let $\overline{S_n}$ be the group of all the automorphisms $\overline{\sigma}$ for $\sigma \in S_n$. Observe that $\overline{S_n}$ is naturally isomorphic to S_n. Let K be the subfield of $F(y_1, \cdots, y_n)$ which is the fixed field of $\overline{S_n}$. Consider the polynomial

$$f(x)\prod_{i=1}^{n}(x - y_i);$$

this polynomial $f(x) \in (F(y_1, \cdots, y_n))[x]$ is a **general polynomial of degree** n. Let $\overline{\sigma_x}$ be the extension of $\overline{\sigma}$, in the natural way, to $(F(y_1, \cdots, y_n))[x]$, where $\overline{\sigma_x}(x) = x$. Now $f(x)$ is left fixed by each map $\overline{\sigma_x}$ for $\sigma \in S_n$; that is,

$$\prod_{i=1}^{n}(x - y_i) = \prod_{i=1}^{n}(x - y_{\sigma(i)}).$$

Thus the coefficients of $f(x)$ are in K; they are *elementary symmetric functions* in the $y_1, \cdots, y_n$. As illustration, note that the constant term of $f(x)$ is

$$(-1)^n y_1 y_2 \cdots y_n,$$

the coefficient of x^{n-1} is $-(y_1 + y_2 + \cdots + y_n)$, and so on. These are symmetric functions in $y_1, \cdots, y_n$.

The first elementary symmetric function in $y_1, \cdots, y_n$ is

$$s_1 = y_1 + y_2 + \cdots + y_n,$$

the second is $s_2 = y_1y_2 + y_1y_3 + \cdots + y_{n-1}y_n$, and so on, and the nth is $s_n = y_1y_2 \cdots y_n$.

Consider the field $E = F(s_1, \cdots, s_n)$. Of course, $E \leq K$, where K is the field of all symmetric functions in $y_1, \cdots, y_n$ over F. But $F(y_1, \cdots, y_n)$ is a finite normal extension

of E, namely, the splitting field of

$$f(x) = \prod_{i=1}^{n}(x - y_i)$$

over E. Since the degree of $f(x)$ is n, we have at once

$$[F(y_1, \cdots, y_n) : E] \leq n!$$

(see Exercise 13, Section 50). However, since K is the fixed field of $\overline{S_n}$ and

$$|\overline{S_n}| = |S_n| = n!,$$

we have also

$$n! \leq \{F(y_1, \cdots, y_n) : K\} \leq [F(y_1, \cdots, y_n) : K].$$

Therefore,

$$n! \leq [F(y_1, \cdots, y_n) : K] \leq [F(y_1, \cdots, y_n) : E] \leq n!,$$

so

$$K = E.$$

The full Galois group of $F(y_1, \cdots, y_n)$ over E is therefore $\overline{S_n}$. The fact that $K = E$ shows that every symmetric function can be expressed as a rational function of the elementary symmetric functions $s_1, \cdots, s_n$. We summarize these results in a theorem.

54.2 Theorem Let $s_1, \cdots, s_n$ be the elementary symmetric functions in the indeterminates $y_1, \cdots, y_n$. Then every symmetric function of $y_1, \cdots, y_n$ over F is a rational function of the elementary symmetric functions. Also, $F(y_1, \cdots, y_n)$ is a finite normal extension of degree $n!$ of $F(s_1, \cdots, s_n)$, and the Galois group of this extension is naturally isomorphic to S_n.

In view of Cayley's Theorem 8.16, it can be deduced from Theorem 54.2 that any finite group can occur as a Galois group (up to isomorphism). (See Exercise 11.)

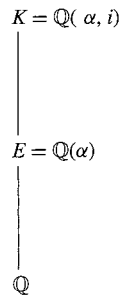
Examples

Let us give our promised example of a finite normal extension having a Galois group whose subgroup diagram does not look like its own inversion.

54.3 Example Consider the splitting field in $\mathbb{C}$ of $x^4 - 2$ over $\mathbb{Q}$. Now $x^4 - 2$ is irreducible over $\mathbb{Q}$, by Eisenstein's criterion, with $p = 2$. Let $\alpha = \sqrt[4]{2}$ be the real positive zero of $x^4 - 2$. Then the four zeros of $x^4 - 2$ in $\mathbb{C}$ are α, $-\alpha$, $i\alpha$, and $-i\alpha$, where i is the usual zero of $x^2 + 1$ in $\mathbb{C}$. The splitting field K of $x^4 - 2$ over $\mathbb{Q}$ thus contains $(i\alpha)/\alpha = i$. Since α is a real number, $\mathbb{Q}(\alpha) < \mathbb{R}$, so $\mathbb{Q}(\alpha) \neq K$. However, since $\mathbb{Q}(\alpha, i)$ contains all zeros of $x^4 - 2$, we see that $\mathbb{Q}(\alpha, i) = K$. Letting $E = \mathbb{Q}(\alpha)$, we have the diagram in Fig. 54.4.

Now $\{1, \alpha, \alpha^2, \alpha^3\}$ is a basis for E over $\mathbb{Q}$, and $\{1, i\}$ is a basis for K over E. Thus

$$\{1, \alpha, \alpha^2, \alpha^3, i, i\alpha, i\alpha^2, i\alpha^3\}$$



54.4 Figure

is a basis for K over $\mathbb{Q}$. Since $[K : \mathbb{Q}] = 8$, we must have $|G(K/\mathbb{Q})| = 8$, so we need to find eight automorphisms of K leaving $\mathbb{Q}$ fixed. We know that any such automorphism σ is completely determined by its values on elements of the basis $\{1, \alpha, \alpha^2, \alpha^3, i, i\alpha, i\alpha^2, i\alpha^3\}$, and these values are in turn determined by $\sigma(\alpha)$ and $\sigma(i)$. But $\sigma(\alpha)$ must always be a conjugate of α over $\mathbb{Q}$, that is, one of the four zeros of $\text{irr}(\alpha, \mathbb{Q}) = x^4 - 2$. Likewise, $\sigma(i)$ must be a zero of $\text{irr}(i, \mathbb{Q}) = x^2 + 1$. Thus the four possibilities for $\sigma(\alpha)$, combined with the two possibilities for $\sigma(i)$, must give all eight automorphisms. We describe these in Table 54.5. For example, $\rho_3(\alpha) = -i\alpha$ and $\rho_3(i) = i$, while ρ_0 is the identity automorphism. Now

$$(\mu_1\rho_1)(\alpha) = \mu_1(\rho_1(\alpha)) = \mu_1(i\alpha) = \mu_1(i)\mu_1(\alpha) = -i\alpha,$$

and, similarly,

$$(\mu_1\rho_1)(i) = -i,$$

so $\mu_1\rho_1 = \delta_2$. A similar computation shows that

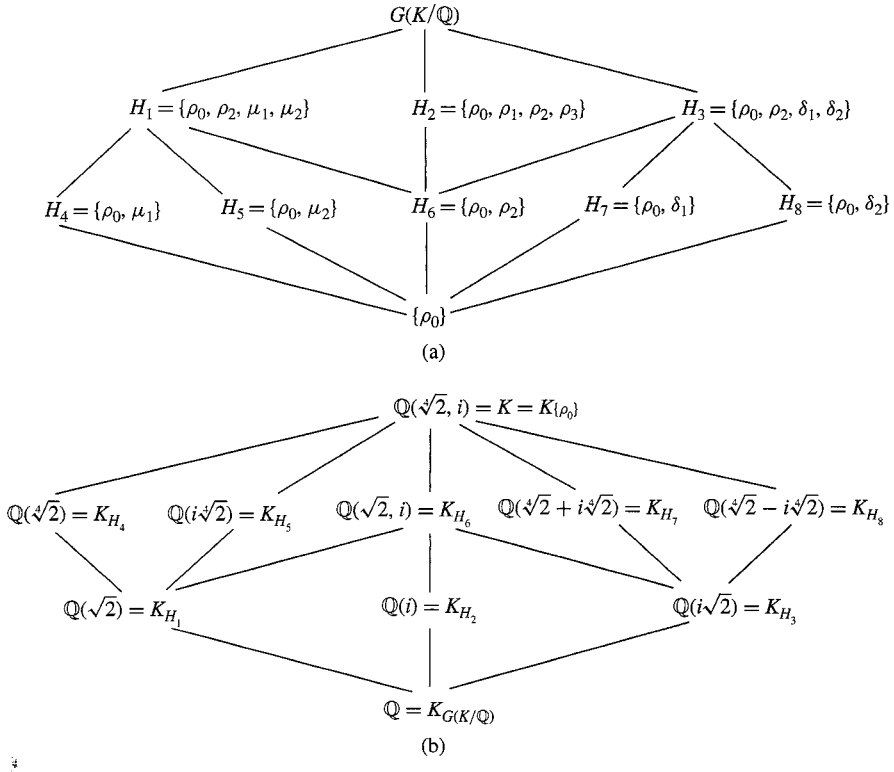
$$(\rho_1\mu_1)(\alpha) = i\alpha \qquad \text{and} \qquad (\rho_1\mu_1)(i) = -i.$$

Thus $\rho_1\mu_1 = \delta_1$, so $\rho_1\mu_1 \neq \mu_1\rho_1$ and $G(K/\mathbb{Q})$ is not abelian. Therefore, $G(K/\mathbb{Q})$ must be isomorphic to one of the two nonabelian groups of order 8 described in Example 40.6. Computing from Table 54.5, we see that ρ_1 is of order 4, μ_1 is of order 2, $\{\rho_1, \mu_1\}$ generates $G(K/\mathbb{Q})$, and $\rho_1\mu_1 = \mu_1\rho_1^3 = \delta_1$. Thus $G(K/\mathbb{Q})$ is isomorphic to the group G_1 of Example 40.6, the *octic group*. We chose our notation for the elements of $G(K/\mathbb{Q})$ so that its group table would coincide with the table for the octic group in Table 8.12. The diagram of subgroups H_i of $G(K/\mathbb{Q})$ is that given in Fig. 8.13. We repeat it here in Fig. 54.6 and also give the corresponding diagram of intermediate fields between $\mathbb{Q}$ and K. This finally illustrates nicely that one diagram is the inversion of the other.

The determination of the fixed fields K_{H_i} sometimes requires a bit of ingenuity. Let's illustrate. To find K_{H_2}, we merely have to find an extension of $\mathbb{Q}$ of degree 2 left fixed by $\{\rho_0, \rho_1, \rho_2, \rho_3\}$. Since all ρ_j leave i fixed, $\mathbb{Q}(i)$ is the field we are after. To find K_{H_4}, we have to find an extension of $\mathbb{Q}$ of degree 4 left fixed by ρ_0 and μ_1. Since μ_1 leaves α fixed and α is a zero of $\text{irr}(\alpha, \mathbb{Q}) = x^4 - 2$, we see that $\mathbb{Q}(\alpha)$ is of degree 4 over $\mathbb{Q}$ and is left fixed by $\{\rho_0, \mu_1\}$. By *Galois theory, it is the only such field.* Here we are using strongly the one-to-one correspondence given by the Galois theory. If we find one field that fits the bill, it is the one we are after. Finding K_{H_7} requires more ingenuity. Since $H_7 = \{\rho_0, \delta_1\}$ is a group, for any $\beta \in K$ we see that $\rho_0(\beta) + \delta_1(\beta)$ is left fixed by ρ_0 and δ_1. Taking $\beta = \alpha$, we see that $\rho_0(\alpha) + \delta_1(\alpha) = \alpha + i\alpha$ is left fixed by H_7. We can check and see that ρ_0 and δ_1 are the only automorphisms leaving $\alpha + i\alpha$ fixed. Thus

54.5 Table

	ρ_0	ρ_1	ρ_2	ρ_3	μ_1	δ_1	μ_2	δ_2
$\alpha \to$	α	$i\alpha$	$-\alpha$	$-i\alpha$	α	$i\alpha$	$-\alpha$	$-i\alpha$
$i \to$	i	i	i	i	$-i$	$-i$	$-i$	$-i$



54.6 Figure (a) Group diagram. (b) Field diagram.

by the one-to-one correspondence, we must have

$$\mathbb{Q}(\alpha + i\alpha) = \mathbb{Q}(\sqrt[4]{2} + i\sqrt[4]{2}) = K_{H_7}.$$

Suppose we wish to find $\text{irr}(\alpha + i\alpha, \mathbb{Q})$. If $\gamma = \alpha + i\alpha$, then for every conjugate of γ over $\mathbb{Q}$, there exists an automorphism of K mapping γ into that conjugate. Thus we need only compute the various different values $\sigma(\gamma)$ for $\sigma \in G(K/\mathbb{Q})$ to find the other zeros of $\text{irr}(\gamma, \mathbb{Q})$. By Theorem 53.2, elements σ of $G(K/\mathbb{Q})$ giving these different values can be found by taking a set of representatives of the left cosets of $G(K/\mathbb{Q}(\gamma)) = \{\rho_0, \delta_1\}$ in $G(K/\mathbb{Q})$. A set of representatives for these left cosets is

$$\{\rho_0, \rho_1, \rho_2, \rho_3\}.$$

The conjugates of $\gamma = \alpha + i\alpha$ are thus $\alpha + i\alpha$, $i\alpha - \alpha$, $-\alpha - i\alpha$, and $-i\alpha + \alpha$. Hence

$$\begin{aligned}\text{irr}(\gamma, \mathbb{Q}) &= [(x - (\alpha + i\alpha))(x - (i\alpha - \alpha))] \\ &\quad \cdot[(x - (-\alpha - i\alpha))(x - (-i\alpha + \alpha))] \\ &= (x^2 - 2i\alpha x - 2\alpha^2)(x^2 + 2i\alpha x - 2\alpha^2) \\ &= x^4 + 4\alpha^4 = x^4 + 8.\end{aligned}$$

▲

We have seen examples in which the splitting field of a quartic (4th degree) polynomial over a field F is an extension of F of degree 8 (Example 54.3) and of degree 24 (Theorem 54.2, with $n = 4$). The degree of an extension of a field F that is a splitting field of a quartic over F must always divide $4! = 24$. The splitting field of $(x-2)^4$ over $\mathbb{Q}$ is $\mathbb{Q}$, an extension of degree 1, and the splitting field of $(x^2-2)^2$ over $\mathbb{Q}$ is $\mathbb{Q}(\sqrt{2})$, an extension of degree 2. Our last example will give an extension of degree 4 for the splitting field of a quartic.

54.7 Example Consider the splitting field of x^4+1 over $\mathbb{Q}$. By Theorem 23.11, we can show that x^4+1 is irreducible over $\mathbb{Q}$, by arguing that it does not factor in $\mathbb{Z}[x]$. (See Exercise 1.) The work on complex number in Section 1 shows that the zeros of x^4+1 are $(1 \pm i)/\sqrt{2}$ and $(-1 \pm i)/\sqrt{2}$. A computation shows that if

$$\alpha = \frac{1+i}{\sqrt{2}},$$

then

$$\alpha^3 = \frac{-1+i}{\sqrt{2}}, \qquad \alpha^5 = \frac{-1-i}{\sqrt{2}}, \qquad \text{and} \qquad \alpha^7 = \frac{1-i}{\sqrt{2}}.$$

Thus the splitting field K of x^4+1 over $\mathbb{Q}$ is $\mathbb{Q}(\alpha)$, and $[K:\mathbb{Q}] = 4$. Let us compute $G(K/\mathbb{Q})$ and give the group and field diagrams. Since there exist automorphisms of K mapping α onto each conjugate of α, and since an automorphism σ of $\mathbb{Q}(\alpha)$ is completely determined by $\sigma(\alpha)$, we see that the four elements of $G(K/\mathbb{Q})$ are defined by Table 54.8. Since*

$$(\sigma_j\sigma_k)(\alpha) = \sigma_j(\alpha^k) = (\alpha^j)^k = \alpha^{jk}$$

and $\alpha^8 = 1$, we see that $G(K/\mathbb{Q})$ is isomorphic to the group $\{1, 3, 5, 7\}$ under multiplication modulo 8. This is the group G_8 of Theorem. 20.6. Since $\sigma_j^2 = \sigma_1$, the identity, for all j, $G(K/\mathbb{Q})$ must be isomorphic to the Klein 4-group. The diagrams are given in Fig. 54.9.

To find $K_{\{\sigma_1,\sigma_3\}}$, it is only necessary to find an element of K not in $\mathbb{Q}$ left fixed by $\{\sigma_1, \sigma_3\}$, since $[K_{\{\sigma_1,\sigma_3\}}:\mathbb{Q}] = 2$. Clearly $\sigma_1(\alpha) + \sigma_3(\alpha)$ is left fixed by both σ_1 and σ_3, since $\{\sigma_1, \sigma_3\}$ is a group. We have

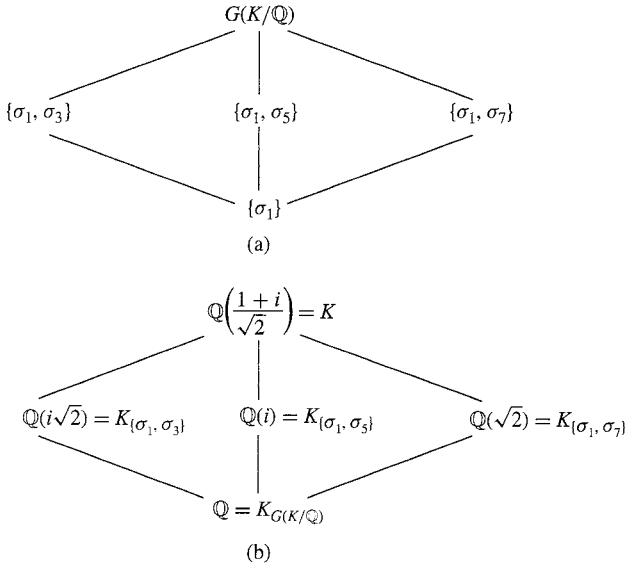
$$\sigma_1(\alpha) + \sigma_3(\alpha) = \alpha + \alpha^3 = i\sqrt{2}.$$

Similarly,

$$\sigma_1(\alpha) + \sigma_7(\alpha) = \alpha + \alpha^7 = \sqrt{2}$$

54.8 Table

	σ_1	σ_3	σ_5	σ_7
$\alpha \to$	α	α^3	α^5	α^7



54.9 Figure (a) Group diagram. (b) Field diagram.

is left fixed by $\{\sigma_1, \sigma_7\}$. This technique is of no use in finding $E_{\{\sigma_1,\sigma_5\}}$, for

$$\sigma_1(\alpha) + \sigma_5(\alpha) = \alpha + \alpha^5 = 0,$$

and $0 \in \mathbb{Q}$. But by a similar argument, $\sigma_1(\alpha)\sigma_5(\alpha)$ is left fixed by both σ_1 and σ_5, and

$$\sigma_1(\alpha)\sigma_5(\alpha) = \alpha\alpha^5 = -i.$$

Thus $\mathbb{Q}(-i) = \mathbb{Q}(i)$ is the field we are after. ▲

EXERCISES 54

Computations (requiring more than the usual amount of theory)

1. Show that $x^4 + 1$ is irreducible in $\mathbb{Q}[x]$, as we asserted in Example 54.7.
2. Verify that the intermediate fields given in the field diagram in Fig. 54.6 are correct (Some are verified in the text. Verify the rest.)
3. For each field in the field diagram in Fig. 54.6, find a primitive element generating the field over $\mathbb{Q}$ (see Theorem 51.15 and give its irreducible polynomial over $\mathbb{Q}$.
4. Let ζ be a primitive 5th root of unity in $\mathbb{C}$.
   a. Show that $\mathbb{Q}(\zeta)$ is the splitting field of $x^5 - 1$ over $\mathbb{Q}$.
   b. Show that every automorphism of $K = \mathbb{Q}(\zeta)$ maps ζ onto some power ζ^r of ζ.
   c. Using part (b), describe the elements of $G(K/\mathbb{Q})$.
   d. Give the group and field diagrams for $\mathbb{Q}(\zeta)$ over $\mathbb{Q}$, computing the intermediate fields as we did in Examples 54.3 and 54.7.

5. Describe the group of the polynomial $(x^5 - 2) \in (\mathbb{Q}(\zeta))[x]$ over $\mathbb{Q}(\zeta)$, where ζ is a primitive 5th root of unity.
6. Repcat Exercise 4 for ζ a primitive 7th root of unity in $\mathbb{C}$.
7. In the easiest way possible, describe the group of the polynomial

$$(x^8 - 1) \in \mathbb{Q}[x]$$

over $\mathbb{Q}$.

8. Find the splitting field K in $\mathbb{C}$ of the polynomial $(x^4 - 4x^2 - 1) \in \mathbb{Q}[x]$. Compute the group of the polynomial over $\mathbb{Q}$ and exhibit the correspondence between the subgroups of $G(K/\mathbb{Q})$ and the intermediate fields. In other words, do the complete job.
9. Express each of the following symmetric functions in y_1, y_2, y_3 over $\mathbb{Q}$ as a rational function of the elementary symmetric functions s_1, s_2, s_3.
   a. ${y_1}^2 + {y_2}^2 + {y_3}^2$
   b. $\dfrac{y_1}{y_2} + \dfrac{y_2}{y_1} + \dfrac{y_1}{y_3} + \dfrac{y_3}{y_1} + \dfrac{y_2}{y_3} + \dfrac{y_3}{y_2}$
10. Let $\alpha_1, \alpha_2, \alpha_3$ be the zeors in $\mathbb{C}$ of the polynomial

$$(x^3 - 4x^2 + 6x - 2) \in \mathbb{Q}[x].$$

Find the polynomial having as zeros precisely the following:

   a. $\alpha_1 + \alpha_2 + \alpha_3$
   b. ${\alpha_1}^2, {\alpha_2}^2, {\alpha_3}^2$

Theory

11. Show that every finite group is isomorphic to some Galois group $G(K/F)$ for some finite normal extension K of some field F.
12. Let $f(x) \in F[x]$ be a monic polynomial of degree n having all its irreducible factors separable over F. Let $K \leq \bar{F}$ be the splitting field of $f(x)$ over F, and suppose that $f(x)$ factors in $K[x]$ into

$$\prod_{i=1}^{n}(x - \alpha_i).$$

Let

$$\Delta(f) = \prod_{i<j}(\alpha_i - \alpha_j);$$

the product $(\Delta(f))^2$ is the **discriminant of** $f(x)$.

   a. Show that $\Delta(f) = 0$ if and only if $f(x)$ has as a factor the square of some irreducible polynomial in $F[x]$.
   b. Show that $(\Delta(f))^2 \in F$.
   c. $G(K/F)$ may be viewed as a subgroup of $\overline{S_n}$, where $\overline{S_n}$ is the group of all permutations of $\{\alpha_i \mid i = 1, \cdots, n\}$. Show that $G(K/F)$, when viewed in this fashion, is a subgroup of $\overline{A_n}$, the group formed by all even permutations of $\{\alpha_i \mid i = 1, \cdots, n\}$, if and only if $\Delta(f) \in F$.

13. An element of $\mathbb{C}$ is an **algebraic integer** if it is a zero of some *monic* polynomial in $\mathbb{Z}[x]$. Show that the set of all algebraic integers forms a subring of $\mathbb{C}$.

SECTION 55 CYCLOTOMIC EXTENSIONS

The Galois Group of a Cyclotomic Extension

This section deals with extension fields of a field F obtained by adjoining to F some roots of unity. The case of a finite field F was covered in Section 33, so we shall be primarily concerned with the case where F is infinite.

55.1 Definition The splitting field of $x^n - 1$ over F is the ***n*th cyclotomic extension of** F. ■

Suppose that F is any field, and consider $(x^n - 1) \in F[x]$. By long division, as in the proof of Lemma 33.8, we see that if α is a zero of $x^n - 1$ and $g(x) = (x^n - 1)/(x - \alpha)$, then $g(\alpha) = (n \cdot 1)(1/\alpha) \neq 0$, provided that the characteristic of F does not divide n. Therefore, under this condition, the splitting field of $x^n - 1$ is a separable and thus a normal extension of F.

■ HISTORICAL NOTE

Carl Gauss considered cyclotomic polynomials in the final chapter of his *Disquisitiones Arithmeticae* of 1801. In that chapter, he gave a constructive procedure for actually determining the roots of $\Phi_p(x)$ in the case where p is prime. Gauss's method, which became an important example for Galois in the development of the general theory, was to solve a series of auxiliary equations, each of degree a prime factor of $p - 1$, with the coefficients of each in turn being determined by the roots of the previous equation. Gauss, of course, knew that the roots of $\Phi_p(x)$ were all powers of one of them, say ζ. He determined the auxiliary equations by taking certain sets of sums of the roots ζ^j, which were the desired roots of these equations. For example, in the case where $p = 19$ (and $p - 1 = 18 = 3 \times 3 \times 2$), Gauss needed to find two equations of degree 3 and one of degree 2 as his auxiliaries. It turned out that the first one had the three roots, $\alpha_1 = \zeta + \zeta^8 + \zeta^7 + \zeta^{18} + \zeta^{11} + \zeta^{12}$, $\alpha_2 = \zeta^2 + \zeta^{16} + \zeta^{14} + \zeta^{17} + \zeta^3 + \zeta^5$, and $\alpha_3 = \zeta^4, + \zeta^{13} + \zeta^9 + \zeta^{15} + \zeta^6 + \zeta^{10}$. In fact, these three values are the roots of the cubic equation $x^3 + x^2 - 6x - 7$. Gauss then found a second cubic equation, with coefficients involving the α's, whose roots were sums of two of the powers of ζ, and finally a quadratic equation, whose coefficients involved the roots of the previous equation, which had ζ as one of its roots. Gauss then asserted (without a complete proof) that each auxiliary equation can in turn be reduced to an equation of the form $x^m - A$, which clearly can be solved by radicals. That is, he showed that the solvability of the Galois group in this case, the cyclic group of order $p - 1$, implied that the cyclotomic equation was solvable in terms of radicals. (See Section 56.)

Assume from now on that this is the case, and let K be the splitting field of $x^n - 1$ over F. Then $x^n - 1$ has n distinct zeros in K, and by Corollary 23.6, these form a cyclic group of order n under the field multiplication. We saw in Corollary 6.16 that a cyclic group of order n has $\varphi(n)$ generators, where φ is the Euler phi-function introduced prior to Theorem 20.8. For our situation here, these $\varphi(n)$ generators are exactly the primitive nth roots of unity.

55.2 Definition The polynomial

$$\Phi_n(x) = \prod_{i=1}^{\varphi(n)} (x - \alpha_i)$$

where the α_i are the primitive nth roots of unity in $\bar{F}$, is the ***n*th cyclotomic polynomial over** F. ■

Since an automorphism of the Galois group $G(K/F)$ must permute the primitive nth roots of unity, we see that $\Phi_n(x)$ is left fixed under every element of $G(K/F)$ regarded as extended in the natural way to $K[x]$. Thus $\Phi_n(x) \in F[x]$. In particular, for $F = \mathbb{Q}$, $\Phi_n(x) \in \mathbb{Q}[x]$, and $\Phi_n(x)$ is a divisor of $x^n - 1$. Thus over $\mathbb{Q}$, we must actually have $\Phi_n(x) \in \mathbb{Z}[x]$, by Theorem 23.11. We have seen that $\Phi_p(x)$ is irreducible over $\mathbb{Q}$, in Corollary 23.17. While $\Phi_n(x)$ need not be irreducible in the case of the fields $\mathbb{Z}_p$, it can be shown that over $\mathbb{Q}$, $\Phi_n(x)$ is irreducible.

Let us now limit our discussion to characteristic 0, in particular to subfields of the complex numbers. Let i be the usual complex zero of $x^2 + 1$. Our work with complex numbers in Section 1 shows that

$$\left(\cos\frac{2\pi}{n} + i\sin\frac{2\pi}{n}\right)^n = \cos 2\pi + i\sin 2\pi = 1,$$

so $\cos(2\pi/n) + i\sin(2\pi/n)$ is an nth root of unity. The least integer m such that $(\cos(2\pi/n) + i\sin(2\pi/n))^m = 1$ is n. *Thus* $\cos(2\pi/n) + i\sin(2\pi/n)$ *is a primitive* n*th root of unity, a zero of*

$$\Phi_n(x) \in \mathbb{Q}[x].$$

55.3 Example A primitive 8th root of unity in $\mathbb{C}$ is

$$\begin{aligned}\zeta &= \cos\frac{2\pi}{8} + i\sin\frac{2\pi}{8} \\ &= \cos\frac{\pi}{4} + i\sin\frac{\pi}{4} \\ &= \frac{1}{\sqrt{2}} + i\frac{1}{\sqrt{2}} = \frac{1+i}{\sqrt{2}}.\end{aligned}$$

By the theory of cyclic groups, in particular by Corollary 6.16 all the primitive 8th roots of unity in $\mathbb{Q}$ are ζ, ζ^3, ζ^5, and ζ^7, so

$$\Phi_8(x) = (x - \zeta)(x - \zeta^3)(x - \zeta^5)(x - \zeta^7).$$

We can compute, directly from this expression, $\Phi_8(x) = x^4 + 1$ (see Exercise 1). Compare this with Example 54.7. ▲

Let us still restrict our work to $F = \mathbb{Q}$, and let us assume, without proof, that $\Phi_n(x)$ is irreducible over $\mathbb{Q}$. Let

$$\zeta = \cos\frac{2\pi}{n} + i\sin\frac{2\pi}{n},$$

so that ζ is a primitive nth root of unity. Note that ζ is a generator of the cyclic multiplicative group of order n consisting of *all* nth roots of unity. All the primitive nth roots of unity, that is, all the generators of this group, are of the form ζ^m for $1 \leq m < n$ and m relatively prime to n. The field $\mathbb{Q}(\zeta)$ is the whole splitting field of $x^n - 1$ over $\mathbb{Q}$. Let $K = \mathbb{Q}(\zeta)$. If ζ^m is another primitive nth root of unity, then since ζ and ζ^m are conjugate over $\mathbb{Q}$, there is an automorphism τ_m in $G(K/\mathbb{Q})$ mapping ζ onto ζ^m. Let τ_r be the similar automorphism in $G(K/\mathbb{Q})$ corresponding to a primitive nth root of unity ζ^r. Then

$$(\tau_m \tau_r)(\zeta) = \tau_m(\zeta^r) = (\tau_m(\zeta))^r = (\zeta^m)^r = \zeta^{rm}.$$

This shows that the Galois group $G(K/\mathbb{Q})$ is isomorphic to the group G_n of Theorem 20.6 consisting of elements of $\mathbb{Z}_n$ relatively prime to n under multiplication modulo n. This group has $\varphi(n)$ elements and is abelian.

Special cases of this material have appeared several times in the text and exercises. For example, α of Example 54.7 is a primitive 8th root of unity, and we made arguments in that example identical to those given here. We summarize these results in a theorem.

55.4 Theorem The Galois group of the nth cyclotomic extension of $\mathbb{Q}$ has $\varphi(n)$ elements and is isomorphic to the group consisting of the positive integers less than n and relatively prime to n under multiplication modulo n.

55.5 Example Example 54.7 illustrates this theorem, for it is easy to see that the splitting field of $x^4 + 1$ is the same as the splitting field of $x^8 - 1$ over $\mathbb{Q}$. This follows from the fact that $\Phi_8(x) = x^4 + 1$ (see Example 55.3 and Exercise 1). ▲

55.6 Corollary The Galois group of the pth cyclotomic extension of $\mathbb{Q}$ for a prime p is cyclic of order $p - 1$.

Proof By Theorem 55.4, the Galois group of the pth cyclotomic extension of $\mathbb{Q}$ has $\varphi(p) = p - 1$ elements, and is isomorphic to the group of positive integers less than p and relatively prime to p under multiplication modulo p. This is exactly the multiplicative group $\langle \mathbb{Z}_p^*, \cdot \rangle$ of nonzero elements of the field $\mathbb{Z}_p$ under field multiplication. By Corollary 23.6, this group is cyclic. ◆

Constructible Polygons

We conclude with an application determining which regular n-gons are constructible with a compass and a straightedge. We saw in Section 32 that the regular n-gon is constructible if and only if $\cos(2\pi/n)$ is a constructible real number. Now let

$$\zeta = \cos\frac{2\pi}{n} + i \sin\frac{2\pi}{n}.$$

Then

$$\frac{1}{\zeta} = \cos\frac{2\pi}{n} - i \sin\frac{2\pi}{n},$$

for

$$\left(\cos\frac{2\pi}{n}+i\sin\frac{2\pi}{n}\right)\left(\cos\frac{2\pi}{n}-i\sin\frac{2\pi}{n}\right)=\cos^2\frac{2\pi}{n}+\sin^2\frac{2\pi}{n}=1.$$

But then

$$\zeta+\frac{1}{\zeta}=2\cos\frac{2\pi}{n}.$$

Thus Corollary 32.8 shows that the regular n-gon is constructible only if $\zeta+1/\zeta$ generates an extension of $\mathbb{Q}$ of degree a power of 2.

If K is the splitting field of x^n-1 over $\mathbb{Q}$, then $[K:\mathbb{Q}]=\varphi(n)$, by Theorem 55.4. If $\sigma\in G(K/\mathbb{Q})$ and $\sigma(\zeta)=\zeta^r$, then

$$\begin{aligned}\sigma\left(\zeta+\frac{1}{\zeta}\right)&=\zeta^r+\frac{1}{\zeta^r}\\&=\left(\cos\frac{2\pi r}{n}+i\sin\frac{2\pi r}{n}\right)+\left(\cos\frac{2\pi r}{n}-i\sin\frac{2\pi r}{n}\right)\\&=2\cos\frac{2\pi r}{n}.\end{aligned}$$

But for $1<r<n$, we have $2\cos(2\pi r/n)=2\cos(2\pi/n)$ only in the case that $r=n-1$. Thus the only elements of $G(K/\mathbb{Q})$ carrying $\zeta+1/\zeta$ onto itself are the identity automorphism and the automorphism τ, with $\tau(\zeta)=\zeta^{n-1}=1/\zeta$. This shows that the subgroup of $G(K/\mathbb{Q})$ leaving $\mathbb{Q}(\zeta+1/\zeta)$ fixed is of order 2, so by Galois theory,

$$\left[\mathbb{Q}\left(\zeta+\frac{1}{\zeta}\right):\mathbb{Q}\right]=\frac{\varphi(n)}{2}.$$

Hence the regular n-gon is constructible only if $\varphi(n)/2$, and therefore also $\varphi(n)$, is a power of 2.

It can be shown by elementary arguments in number theory that if

$$n=2^{\nu}{p_1}^{s_1}\cdots{p_t}^{s_t},$$

where the p_i are the distinct odd primes dividing n, then

$$\varphi(n)=2^{\nu-1}{p_1}^{s_1-1}\cdots{p_t}^{s_t-1}(p_1-1)\cdots(p_t-1). \tag{1}$$

If $\varphi(n)$ is to be a power of 2, then every odd prime dividing n must appear only to the first power and must be one more than a power of 2. Thus we must have each

$$p_i=2^m+1$$

for some m. Since -1 is a zero of x^q+1 for q an odd prime, $x+1$ divides x^q+1 for q an odd prime. Thus, if $m=qu$, where q is an odd prime, then $2^m+1=(2^u)^q+1$ is divisible by 2^u+1. Therefore, for $p_i=2^m+1$ to be prime, it must be that m is divisible by 2 only, so p_i has to have the form

$$p_i=2^{(2^k)}+1,$$

a **Fermat prime.** Fermat conjectured that these numbers $2^{(2^k)} + 1$ were prime for all nonnegative integers k. Euler showed that while $k = 0, 1, 2, 3$, and 4 give the primes 3, 5, 17, 257, and 65537, for $k = 5$, the integer $2^{(2^5)} + 1$ is divisible by 641. It has been shown that for $5 \leq k \leq 19$, all the numbers $2^{(2^k)} + 1$ are composite. The case $k = 20$ is still unsolved as far as we know. For at least 60 values of k greater than 20, including $k = 9448$, it has been shown that $2^{2^k} + 1$ is composite. It is unknown whether the number of Fermat primes is finite or infinite.

We have thus shown that the only regular n-gons that might be constructible are those where the odd primes dividing n are Fermat primes whose squares do not divide n. In particular, the only regular p-gons that might be constructible for p a prime greater than 2 are those where p is a Fermat prime.

55.7 Example The regular 7-gon is not constructible, since 7 is not a Fermat prime. Similarly, the regular 18-gon is not constructible, for while 3 is a Fermat prime, its square divides 18. ▲

It is a fact that we now demonstrate that all these regular n-gons that are candidates for being constructible are indeed actually constructible. Let ζ again be the primitive nth root of unity $\cos(2\pi/n) + i \sin(2\pi/n)$. We saw above that

$$2\cos\frac{2\pi}{n} = \zeta + \frac{1}{\zeta},$$

and that

$$\left[\mathbb{Q}\left(\zeta + \frac{1}{\zeta}\right) : \mathbb{Q}\right] = \frac{\varphi(n)}{2}.$$

Suppose now that $\varphi(n)$ is a power 2^s of 2. Let E be $\mathbb{Q}(\zeta + 1/\zeta)$. We saw above that $\mathbb{Q}(\zeta + 1/\zeta)$ is the subfield of $K = \mathbb{Q}(\zeta)$ left fixed by $H_1 = \{\iota, \tau\}$, where ι is the identity element of $G(K/\mathbb{Q})$ and $\tau(\zeta) = 1/\zeta$. By Sylow theory, there exist additional subgroups H_j of order 2^j of $G(\mathbb{Q}(\zeta)/\mathbb{Q})$ for $j = 0, 2, 3, \cdots, s$ such that

$$\{\iota\} = H_0 < H_1 < \cdots < H_s = G(\mathbb{Q}(\zeta)/\mathbb{Q}).$$

By Galois theory,

$$\mathbb{Q} = K_{H_s} < K_{H_{s-1}} < \cdots < K_{H_1} = \mathbb{Q}\left(\zeta + \frac{1}{\zeta}\right),$$

and $[K_{H_{j-1}} : K_H] = 2$. Note that $(\zeta + 1/\zeta) \in \mathbb{R}$, so $\mathbb{Q}(\zeta + 1/\zeta) < \mathbb{R}$. If $K_{H_{j-1}} = K_{H_j}(\alpha_j)$, then α_j is a zero of some $(a_j x^2 + b_j x + c_j) \in K_{H_j}[x]$. By the familiar "quadratic formula," we have

$$K_{H_{j-1}} = K_{H_j}\left(\sqrt{b_j^2 - 4a_j c_j}\right).$$

Since we saw in Section 33 that construction of square roots of positive constructible numbers can be achieved by a straightedge and a compass, we see that every element in

$\mathbb{Q}(\zeta + 1/\zeta)$, in particular $\cos(2\pi/n)$, is constructible. Hence the regular n-gons where $\varphi(n)$ is a power of 2 are constructible.

We summarize our work under this heading in a theorem.

55.8 Theorem The regular n-gon is constructible with a compass and a straightedge if and only if all the odd primes dividing n are Fermat primes whose squares do not divide n.

55.9 Example The regular 60-gon is constructible, since $60 = (2^2)(3)(5)$ and 3 and 5 are both Fermat primes. ▲

EXERCISES 55

Computations

1. Referring to Example 55.3, complete the indicated computation, showing that $\Phi_8(x) = x^4 + 1$. [*Suggestion:* Compute the product in terms of ζ, and then use the fact that $\zeta^8 = 1$ and $\zeta^4 = -1$ to simplify the coefficients.]
2. Classify the group of the polynomial $(x^{20} - 1) \in \mathbb{Q}[x]$ over $\mathbb{Q}$ according to the Fundamental Theorem of finitely generated abelian groups. [*Hint:* Use Theorem 55.4.]
3. Using the formula for $\varphi(n)$ in terms of the factorization of n, as given in Eq. (1), compute the indicated value:

   a. $\varphi(60)$ **b.** $\varphi(1000)$ **c.** $\varphi(8100)$
4. Give the first 30 values of $n \geq 3$ for which the regular n-gon is constructible with a straightedge and a compass.
5. Find the smallest angle of integral degree, that is, 1°, 2°, 3°, and so on, constructible with a straightedge and a compass. [*Hint:* Constructing a 1° angle amounts to constructing the regular 360-gon, and so on.]
6. Let K be the splitting field of $x^{12} - 1$ over $\mathbb{Q}$.

   a. Find $[K : \mathbb{Q}]$.

   b. Show that for $\sigma \in G(K/\mathbb{Q})$, σ^2 is the identity automorphism. Classify $G(K/\mathbb{Q})$ according to the Fundamental Theorem 11.12 of finitely generated abelian groups.
7. Find $\Phi_3(x)$ over $\mathbb{Z}_2$. Find $\Phi_8(x)$ over $\mathbb{Z}_3$.
8. How many elements are there in the splitting field of $x^6 - 1$ over $\mathbb{Z}_3$?

Concepts

9. Mark each of the following true or false.

   _____ **a.** $\Phi_n(x)$ is irreducible over every field of characteristic 0.

   _____ **b.** Every zero in $\mathbb{C}$ of $\Phi_n(x)$ is a primitive nth root of unity.

   _____ **c.** The group of $\Phi_n(x) \in \mathbb{Q}[x]$ over $\mathbb{Q}$ has order n.

   _____ **d.** The group of $\Phi_n(x) \in \mathbb{Q}[x]$ over $\mathbb{Q}$ is abelian.

   _____ **e.** The Galois group of the splitting field of $\Phi_n(x)$ over $\mathbb{Q}$ has order $\varphi(n)$.

   _____ **f.** The regular 25-gon is constructible with a straightedge and a compass.

   _____ **g.** The regular 17-gon is constructible with a straightedge and a compass.

   _____ **h.** For a prime p, the regular p-gon is constructible if and only if p is a Fermat prime.

   _____ **i.** All integers of the form $2^{(2^k)} + 1$ for nonnegative integers k are Fermat primes.

   _____ **j.** All Fermat primes are numbers of the form $2^{(2^k)} + 1$ for nonnegative integers k.

Theory

10. Show that if F is a field of characteristic not dividing n, then

$$x^n - 1 = \prod_{d \mid n} \Phi_d(x)$$

in $F[x]$, where the product is over all divisors d of n.

11. Find the cyclotomic polynomial $\Phi_n(x)$ over $\mathbb{Q}$ for $n = 1, 2, 3, 4, 5$, and 6. [*Hint:* Use Exercise 10.]

12. Find $\Phi_{12}(x)$ in $\mathbb{Q}[x]$. [*Hint:* Use Exercises 10 and 11.]

13. Show that in $\mathbb{Q}[x]$, $\Phi_{2n}(x) = \Phi_n(-x)$ for odd integers $n > 1$. [*Hint:* If ζ is a primitive nth root of unity for n odd, what is the order of $-\zeta$?]

14. Let $n, m \in \mathbb{Z}^+$ be relatively prime. Show that the splitting field in $\mathbb{C}$ of $x^{nm} - 1$ over $\mathbb{Q}$ is the same as the splitting field in $\mathbb{C}$ of $(x^n - 1)(x^m - 1)$ over $\mathbb{Q}$.

15. Let $n, m \in \mathbb{Z}^+$ be relatively prime. Show that the group of $(x^{nm} - 1) \in \mathbb{Q}[x]$ over $\mathbb{Q}$ is isomorphic to the direct product of the groups of $(x^n - 1) \in \mathbb{Q}[x]$ and of $(x^m - 1) \in \mathbb{Q}[x]$ over $\mathbb{Q}$. [*Hint:* Using Galois theory, show that the groups of $x^m - 1$ and $x^n - 1$ can both be regarded as subgroups of the group of $x^{nm} - 1$. Then use Exercises 50 and 51 of Section 11.]

SECTION 56 INSOLVABILITY OF THE QUINTIC

The Problem

We are familiar with the fact that a quadratic polynomial $f(x) = ax^2 + bx + c$, $a \neq 0$, with real coefficients has $(-b \pm \sqrt{b^2 - 4ac})/2a$ as zeros in $\mathbb{C}$. Actually, this is true for $f(x) \in F[x]$, where F is any field of characteristic $\neq 2$ and the zeros are in $\overline{F}$. Exercise 4 asks us to show this. Thus, for example, $(x^2 + 2x + 3) \in \mathbb{Q}[x]$ has its zeros in $\mathbb{Q}(\sqrt{-2})$. You may wonder whether the zeros of a cubic polynomial over $\mathbb{Q}$ can also always be expressed in terms of radicals. The answer is yes, and indeed, even the zeros of a polynomial of degree 4 over $\mathbb{Q}$ can be expressed in terms of radicals. After mathematicians had tried for years to find the "radical formula" for zeros of a 5th degree polynomial, it was a triumph when Abel proved that a quintic need not be solvable by radicals. Our first job will be to describe precisely what this means. A large amount of the algebra we have developed is used in the forthcoming discussion.

Extensions by Radicals

56.1 Definition An extension K of a field F is an **extension of F by radicals** if there are elements $\alpha_1, \cdots, \alpha_r \in K$ and positive integers $n_1, \cdots, n_r$ such that $K = F(\alpha_1, \cdots, \alpha_r)$, $\alpha_1^{n_1} \in F$ and $\alpha_i^{n_i} \in F(\alpha_1, \cdots, \alpha_{i-1})$ for $1 < i \leq r$. A polynomial $f(x) \in F[x]$ is **solvable by radicals over** F if the splitting field E of $f(x)$ over F is contained in an extension of F by radicals. ■

A polynomial $f(x) \in F(x)$ is thus solvable by radicals over F if we can obtain every zero of $f(x)$ by using a finite sequence of the operations of addition, subtraction, multiplication, division, and taking n_ith roots, starting with elements of F. Now to say that the quintic is not solvable in the classic case, that is, characteristic 0, is not to say that no quintic is solvable, as the following example shows.

■ Historical Note

The first publication of a formula for solving cubic equations in terms of radicals was in 1545 in the *Ars Magna* of Girolamo Cardano, although the initial discovery of the method is in part also due to Scipione del Ferro and Niccolo Tartaglia. Cardano's student, Lodovico Ferrari, discovered a method for solving quartic equations by radicals, which also appeared in Cardano's work.

After many mathematicians had attempted to solve quintics by similar methods, it was Joseph-Louis Lagrange who in 1770 first attempted a detailed analysis of the general principles underlying the solutions for polynomials of degree 3 and 4, and showed why these methods fail for those of higher degree. His basic insight was that in the former cases there were rational functions of the roots that took on two and three values, respectively, under all possible permutations of the roots, hence these rational functions could be written as roots of equations of degree less than that of the original. No such functions were evident in equations of higher degree.

The first mathematician to claim to have a proof of the insolvability of the quintic equation was Paolo Ruffini (1765–1822) in his algebra text of 1799. His proof was along the lines suggested by Lagrange, in that he in effect determined all of the subgroups of S_5 and showed how these subgroups acted on rational functions of the roots of the equation. Unfortunately, there were several gaps in his various published versions of the proof. It was Niels Henrik Abel who, in 1824 and 1826, published a complete proof, closing all of Ruffini's gaps and finally settling this centuries-old question.

56.2 Example The polynomial $x^5 - 1$ is solvable by radicals over $\mathbb{Q}$. The splitting field K of $x^5 - 1$ is generated over $\mathbb{Q}$ by a primitive 5th root ζ of unity. Then $\zeta^5 = 1$, and $K = \mathbb{Q}(\zeta)$. Similarly, $x^5 - 2$ is solvable by radicals over $\mathbb{Q}$, for its splitting field over $\mathbb{Q}$ is generated by $\sqrt[5]{2}$ and ζ, where $\sqrt[5]{2}$ is the real zero of $x^5 - 2$. ▲

To say that the quintic is insolvable in the classic case means that there exists *some* polynomial of degree 5 with real coefficients that is not solvable by radicals. We shall show this. *We assume throughout this section that all fields mentioned have characteristic 0.*

The outline of the argument is as follows, and it is worthwhile to try to remember it.

1. *We shall show that a polynomial $f(x) \in F[x]$ is solvable by radicals over F (if and) only if its splitting field E over F has a solvable Galois group.* Recall that a solvable group is one having a composition series with *abelian* quotients. While this theorem goes both ways, we shall not prove the "if" part.
2. *We shall show that there is a subfield F of the real numbers and a polynomial $f(x) \in F[x]$ of degree 5 with a splitting field E over F such that $G(E/F) \simeq S_5$, the symmetric group on 5 letters.* Recall that a composition series for S_5 is $\{\iota\} < A_5 < S_5$. Since A_5 is not abelian, we will be done.

The following lemma does most of our work for Step 1.

56.3 Lemma Let F be a field of characteristic 0, and let $a \in F$. If K is the splitting field of $x^n - a$ over F, then $G(K/F)$ is a solvable group.

Proof Suppose first that F contains all the nth roots of unity. By Corollary 23.6 the nth roots of unity form a cyclic subgroup of $\langle F^*, \cdot \rangle$. Let ζ be a generator of the subgroup. (Actually, the generators are exactly the *primitive* nth roots of unity.) Then the nth roots of unity are

$$1, \zeta, \zeta^2, \cdots, \zeta^{n-1}.$$

If $\beta \in \bar{F}$ is a zero of $(x^n - a) \in F[x]$, then all zeros of $x^n - a$ are

$$\beta, \zeta\beta, \zeta^2\beta, \cdots, \zeta^{n-1}\beta.$$

Since $K = F(\beta)$, an automorphism σ in $G(K/F)$ is determined by the value $\sigma(\beta)$ of the automorphism σ on β. Now if $\sigma(\beta) = \zeta^i\beta$ and $\tau(\beta) = \zeta^j\beta$, where $\tau \in G(K/F)$, then

$$(\tau\sigma)(\beta) = \tau(\sigma(\beta)) = \tau(\zeta^i\beta) = \zeta^i\tau(\beta) = \zeta^i\zeta^j\beta,$$

since $\zeta^i \in F$. Similarly,

$$(\sigma\tau)(\beta) = \zeta^j\zeta^i\beta.$$

Thus $\sigma\tau = \tau\sigma$, and $G(K/F)$ is abelian and therefore solvable.

Now suppose that F does not contain a primitive nth root of unity. Let ζ be a generator of the cyclic group of nth roots of unity under multiplication in $\bar{F}$. Let β again be a zero of $x^n - a$. Since β and $\zeta\beta$ are both in the splitting field K of $x^n - a$, $\zeta = (\zeta\beta)/\beta$ is in K. Let $F' = F(\zeta)$, so we have $F < F' \leq K$. Now F' is a normal extension of F, since F' is the splitting field of $x^n - 1$. Since $F' = F(\zeta)$, an automorphism η in $G(F'/F)$ is determined by $\eta(\zeta)$, and we must have $\eta(\zeta) = \zeta^i$ for some i, since all zeros of $x^n - 1$ are powers of ζ. If $\mu(\zeta) = \zeta^j$ for $\mu \in G(F'/F)$, then

$$(\mu\eta)(\zeta) = \mu(\eta(\zeta)) = \mu(\zeta^i) = \mu(\zeta)^i = (\zeta^j)^i = \zeta^{ij},$$

and, similarly,

$$(\eta\mu)(\zeta) = \zeta^{ij}.$$

Thus $G(F'/F)$ is abelian. By the Main Theorem of Galois Theory,

$$\{\iota\} \leq G(K/F') \leq G(K/F)$$

is a normal series and hence a subnormal series of groups. The first part of the proof shows that $G(K/F')$ is abelian, and Galois theory tells us that $G(K/F)/G(K/F')$ is isomorphic to $G(F'/F)$, which is abelian. Exercise 6 shows that if a group has a subnormal series of subgroups with abelian quotient groups, then any refinement of this series also has abelian quotient groups. Thus a composition series of $G(K/F)$ must have abelian quotient groups, so $G(K/F)$ is solvable. ◆

The following theorem will complete Part 1 of our program.

56.4 Theorem Let F be a field of characteristic zero, and let $F \leq E \leq K \leq \bar{F}$, where E is a normal extension of F and K is an extension of F by radicals. Then $G(E/F)$ is a solvable group.

Proof We first show that K is contained in a finite normal extension L of F by radicals and that the group $G(L/F)$ is solvable. Since K is an extension by radicals, $K = F(\alpha_1, \cdots, \alpha_r)$

where $\alpha_i^{n_i} \in F(\alpha_1, \cdots, \alpha_{i-1})$ for $1 < i \leq r$ and $\alpha_1^{n_1} \in F$. To form L, we first form the splitting field L_1 of $f_1(x) = x^{n_1} - \alpha_1^{n_1}$ over F. Then L_1 is a normal extension of F, and Lemma 56.3 shows that $G(L_1/F)$ is a solvable group. Now $\alpha_2^{n_2} \in L_1$ and we form the polynomial

$$f_2(x) = \prod_{\sigma \in G(L_1/F)} [(x^{n_2} - \sigma(\alpha_2)^{n_2}].$$

Since this polynomial is invariant under action by any σ in $G(L_1/F)$, we see that $f_2(x) \in F[x]$. We let L_2 be the splitting field of $f_2(x)$ over L_1. Then L_2 is a splitting field over F also and is a normal extension of F by radicals. We can form L_2 from L_1 via repeated steps as in Lemma 56.3, passing to a splitting field of $x^{n_2} - \sigma(\alpha_2)^{n_2}$ at each step. By Lemma 56.3 and Exercise 7, we see that the Galois group over F of each new extension thus formed continues to be solvable. We continue this process of forming splitting fields over F in this manner: At stage i, we form the splitting field of the polynomial

$$f_i(x) = \prod_{\alpha \in G(L_{i-1}/F)} [(x^{n_i} - \sigma(\alpha_i)^{n_i}]$$

over L_{i-1}. We finally obtain a field $L = L_r$ that is a normal extension of F by radicals, and we see that $G(L/F)$ is a solvable group. We see from construction that $K \leq L$.

To conclude, we need only note that by Theorem 53.6, we have $G(E/F) \simeq G(L/F)/G(L/E)$. Thus $G(E/F)$ is a factor group, and hence a homomorphic image, of $G(L/F)$. Since $G(L/F)$ is solvable, Exercise 29 of Section 35 shows that $G(E/F)$ is solvable. ◆

The Insolvability of the Quintic

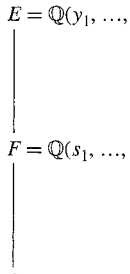
It remains for us to show that there is a subfield F of the real numbers and a polynomial $f(x) \in F[x]$ of degree 5 such that the splitting field E of $f(x)$ over F has a Galois group isomorphic to S_5.

Let $y_1 \in \mathbb{R}$ be transcendental over $\mathbb{Q}$, $y_2 \in \mathbb{R}$ be transcendental over $\mathbb{Q}(y_1)$, and so on, until we get $y_5 \in \mathbb{R}$ transcendental over $\mathbb{Q}(y_1, \cdots, y_4)$. It can be shown by a counting argument that such transcendental real numbers exist. Transcendentals found in this fashion are **independent transcendental elements over** $\mathbb{Q}$. Let $E = \mathbb{Q}(y_1, \cdots, y_5)$, and let

$$f(x) = \prod_{i=1}^{5} (x - y_i).$$

Thus $f(x) \in E[x]$. Now the coefficients of $f(x)$ are, except possibly for sign, among the *elementary symmetry functions* in the y_i, namely

$$\begin{aligned} s_1 &= y_1 + y_2 + \cdots + y_5, \\ s_2 &= y_1y_2 + y_1y_3 + y_1y_4 + y_1y_5 + y_2y_3 \\ &\quad + y_2y_4 + y_2y_5 + y_3y_4 + y_3y_5 + y_4y_5, \\ &\vdots \\ s_5 &= y_1y_2y_3y_4y_5. \end{aligned}$$



56.5 Figure

The coefficient of x^i in $f(x)$ is $\pm s_{5-i}$. Let $F = \mathbb{Q}(s_1, s_2, \cdots, s_5)$; then $f(x) \in F[x]$ (see Fig. 56.5). Then E is the splitting field over F of $f(x)$. Since the y_i behave as indeterminates over $\mathbb{Q}$, for each $\sigma \in S_5$, the symmetric group on five letters, σ induces an automorphism $\bar{\sigma}$ of E defined by $\bar{\sigma}(a) = a$ for $a \in \mathbb{Q}$ and $\bar{\sigma}(y_i) = y_{\sigma(i)}$. Since $\prod_{i=1}^{5}(x - y_i)$ is the same polynomial as $\prod_{i=1}^{5}(x - y_{\sigma(i)})$, we have

$$\bar{\sigma}(s_i) = s_i$$

for each i, so $\bar{\sigma}$ leaves F fixed, and hence $\bar{\sigma} \in G(E/F)$. Now S_5 has order 5!, so

$$|G(E/F)| \geq 5!.$$

Since the splitting field of a polynomial of degree 5 over F has degree at most 5! over F, we see that

$$|G(E/F)| \leq 5!.$$

Thus $|G(E/F)| = 5!$, and the automorphisms $\bar{\sigma}$ make up the full Galois group $G(E/F)$. Therefore, $G(E/F) \simeq S_5$, so $G(E/F)$ is not solvable. This completes our outline, and we summarize in a theorem.

56.6 Theorem Let $y_1, \cdots, y_5$ be independent transcendental real numbers over $\mathbb{Q}$. The polynomial

$$f(x) = \prod_{i=1}^{5}(x - y_i)$$

is not solvable by radicals over $F = \mathbb{Q}(s_1, \cdots, s_5)$, where s_i is the ith elementary symmetric function in $y_1, \cdots, y_5$.

It is evident that a generalization of these arguments shows that (*final goal*) a polynomial of degree n need not be solvable by radicals for $n \geq 5$.

In conclusion, we comment that there exist polynomials of degree 5 in $\mathbb{Q}[x]$ that are not solvable by radicals over $\mathbb{Q}$. A demonstration of this is left to the exercises (see Exercise 8).

EXERCISES 56

Concepts

1. Can the splitting field K of $x^2 + x + 1$ over $\mathbb{Z}_2$ be obtained by adjoining a square root to $\mathbb{Z}_2$ of an element in $\mathbb{Z}_2$? Is K an extension of $\mathbb{Z}_2$ by radicals?
2. Is every polynomial in $F[x]$ of the form $ax^8 + bx^6 + cx^4 + dx^2 + e$, where $a \neq 0$, solvable by radicals over F, if F is of characteristic 0? Why or why not?
3. Mark each of the following true of false.

   _____ **a.** Let F be a field of characteristic 0. A polynomial in $F[x]$ is solvable by radicals if and only if its splitting field in $\bar{F}$ is contained in an extension of F by radicals.

   _____ **b.** Let F be a field of characteristic 0. A polynomial in $F[x]$ is solvable by radicals if and only if its splitting field in $\bar{F}$ has a solvable Galois group over F.

_____ **c.** The splitting field of $x^{17} - 5$ over $\mathbb{Q}$ has a solvable Galois group.

_____ **d.** The numbers π and $\sqrt{\pi}$ are independent transcendental numbers over $\mathbb{Q}$.

_____ **e.** The Galois group of a finite extension of a finite field is solvable.

_____ **f.** No quintic polynomial is solvable by radicals over any field.

_____ **g.** Every 4th degree polynomial over a field of characteristic 0 is solvable by radicals.

_____ **h.** The zeros of a cubic polynomial over a field F of characteristic 0 can always be attained by means of a finite sequence of operations of addition, subtraction, multiplication, division, and taking square roots starting with elements in F.

_____ **i.** The zeros of a cubic polynomial over a field F of characteristic 0 can never be attained by means of a finite sequence of operations of addition, subtraction, multiplication, division, and taking square roots, starting with elements in F.

_____ **j.** The theory of subnormal series of groups play an important role in applications of Galois theory.

Theory

4. Let F be a field, and let $f(x) = ax^2 + bx + c$ be in $F[x]$, where $a \neq 0$. Show that if the characteristic of F is not 2, the splitting field of $f(x)$ over F is $F(\sqrt{b^2 - 4ac})$. [*Hint:* Complete the square, just as in your high school work, to derive the "quadratic formula."]

5. Show that if F is a field of characteristic different from 2 and

$$f(x) = ax^4 + bx^2 + c,$$

where $a \neq 0$, then $f(x)$ is solvable by radicals over F.

6. Show that for a finite group, every refinement of a subnormal series with abelian quotients also has abelian quotients, thus completing the proof of Lemma 56.3. [*Hint:* Use Theorem 34.7.]

7. Show that for a finite group, a subnormal series with solvable quotient groups can be refined to a composition series with abelian quotients, thus completing the proof of Theorem 56.4. [*Hint:* Use Theorem 34.7.]

8. This exercise exhibits a polynomial of degree 5 in $\mathbb{Q}[x]$ that is not solvable by radicals over $\mathbb{Q}$.

a. Show that if a subgroup H of S_5 contains a cycle of length 5 and a transposition τ, then $H = S_5$. [*Hint:* Show that H contains every transposition of S_5 and apply Corollary 9.12. See Exercise 39, Section 9.]

b. Show that if $f(x)$ is an irreducible polynomial in $\mathbb{Q}[x]$ of degree 5 having exactly two complex and three real zeros in $\mathbb{C}$, then the group of $f(x)$ over $\mathbb{Q}$ is isomorphic to S_5. [*Hint:* Use Sylow theory to show that the group has an element of order 5. Use the fact that $f(x)$ has exactly two complex zeros to show that the group has an element of order 2. Then apply part (a).]

c. The polynomial $f(x) = 2x^5 - 5x^4 + 5$ is irreducible in $\mathbb{Q}[x]$, by the Eisenstein criterion, with $p = 5$. Use the techniques of calculus to find relative maxima and minima and to "graph the polynomial function f" well enough to see that $f(x)$ must have exactly three real zeros in $\mathbb{C}$. Conclude from part (b) and Theorem 56.4 that $f(x)$ is not solvable by radicals over $\mathbb{Q}$.